FPT Education

FPT UNIVERSITY

PART I
IMAGE PROCESSING

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SCHOOL OF ELECTRICAL ENGINEERING, INTERNATIONAL UNIVERSITY (VNU-HCMC)

Ho Chi Minh City, Fall 2025

LECTURE VIII – MORPHOLOGICAL IMAGE PROCESSING

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LECTURE CONTENT

- What is mathematical morphology and how is it used in image processing?
- What are the main morphological operations and what is the effect of applying them to binary and grayscale images?
- What is a structuring element (SE) and how does it impact the result of a morphological operation?
- What are some of the most useful morphological image processing algorithms?
- Chapter Summary – What we have learned?
- Problems

8.1 INTRODUCTION

- Mathematical morphology is a branch of image processing that has been successfully used to *provide tools for representing, describing,* and *analyzing shapes in images*.
- As its emphasis on studying the geometrical structure of the components of an image. In addition to providing useful tools for *extracting image components*, morphological algorithms have been used for pre- or postprocessing the images containing shapes of interest.
- The basic principle of mathematical morphology is the extraction of geometrical and topological information from an unknown set (an image) through transformations using another, well-defined, set known as *Structuring Element*. In morphological image processing, the design of SEs, their shape and size, is crucial to the success of the morphological operations that use them.

8.2 FUNDAMENTAL CONCEPTS AND OPERATIONS

- The basic concepts of mathematical morphology can be introduced with the help of set theory and its standard operations: *union* (∪), *intersection* (∩), and *complement*, defined as

$$A^c = \{z | z \notin A\}$$

and *the difference* of two sets A and B:

$$A - B = \{z | z \in A, z \notin B\} = A \cap B^c$$

Let A be a set (of pixels in a binary image) and $\omega = (x, y)$ be a particular coordinate point. The *translation* of set A by point ω is denoted by A_ω and defined as

$$A_\omega = \{c | c = a + \omega, for\ a \in A\}$$

The reflection of set A relative to the origin of a coordinate system, denoted $\hat{A}$, is defined as

$$\hat{A} = \{z | z = -a, for\ a \in A\}$$

8.2 FUNDAMENTAL CONCEPTS AND OPERATIONS

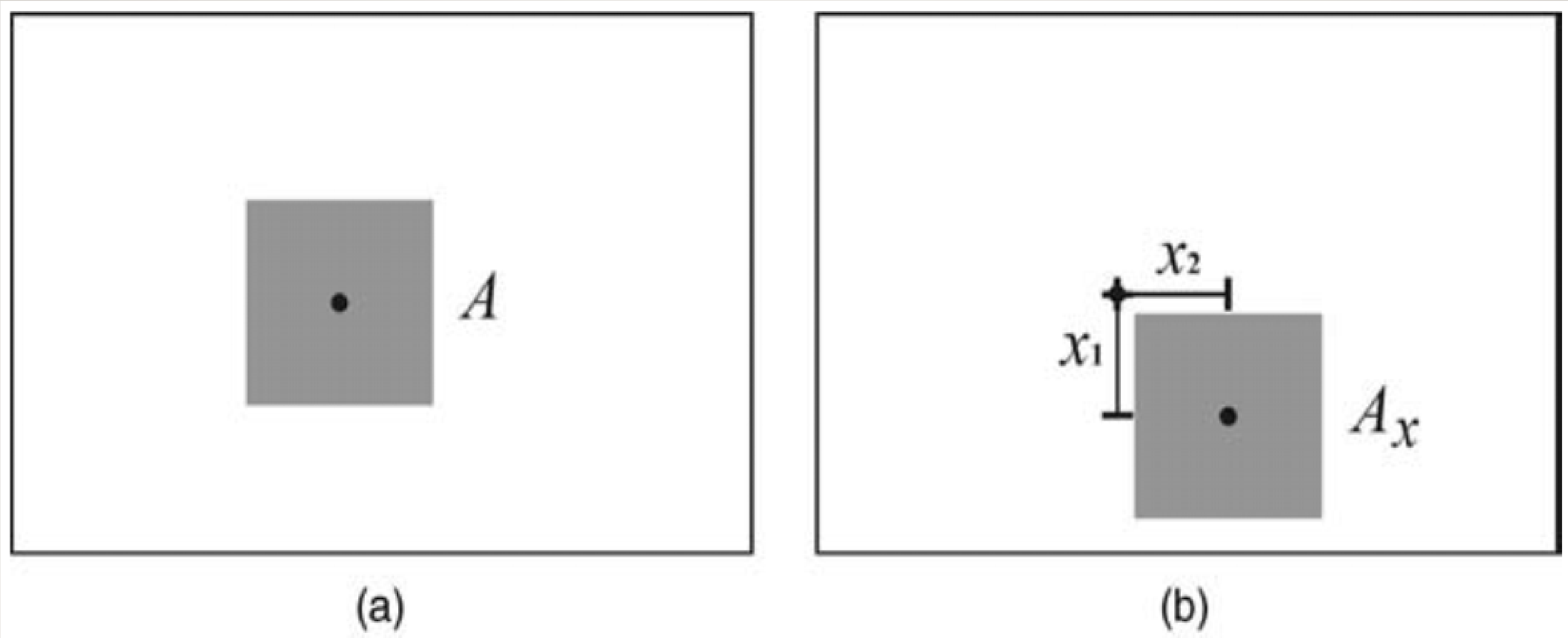


FIGURE 8.1 Basic set operations: (a) set A; (b) translation of A by x = (x_1,x_2);

8.2 FUNDAMENTAL CONCEPTS AND OPERATIONS

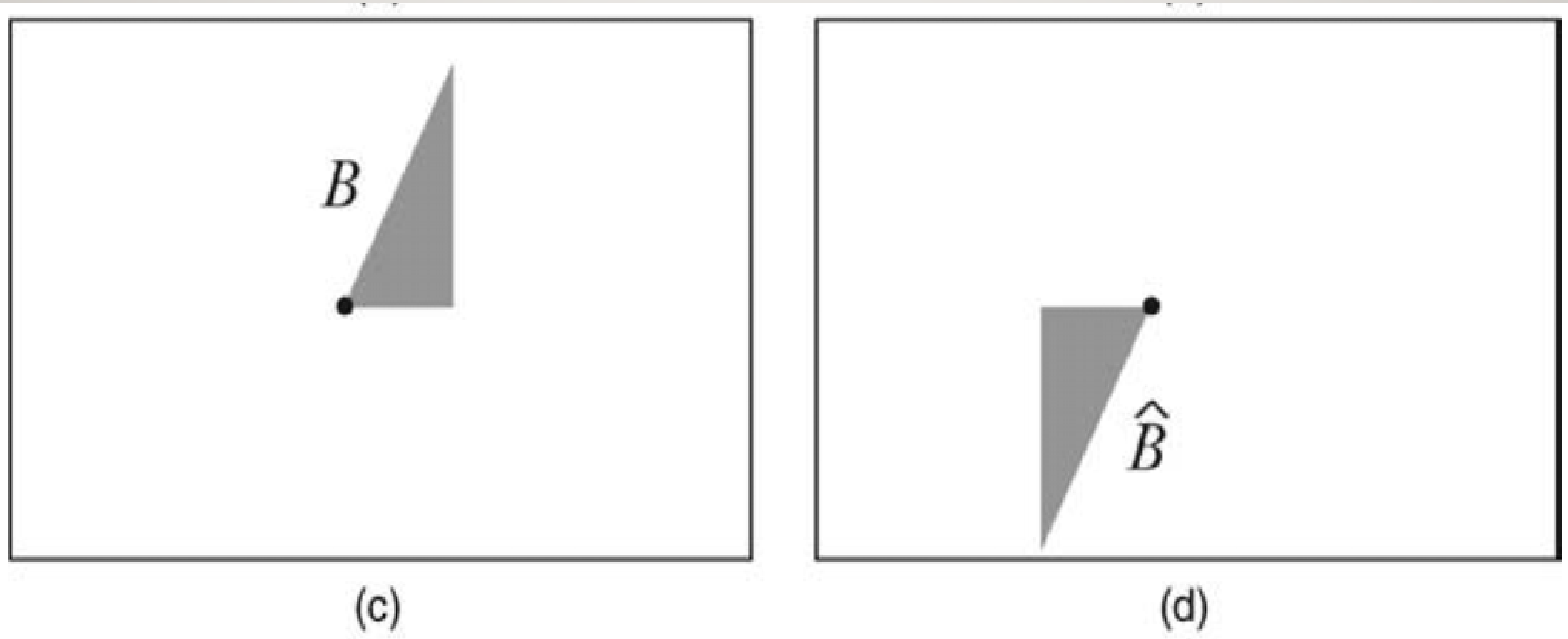


FIGURE 8.1 Basic set operations: (c) set B; (d) reflection of B;

8.2 FUNDAMENTAL CONCEPTS AND OPERATIONS

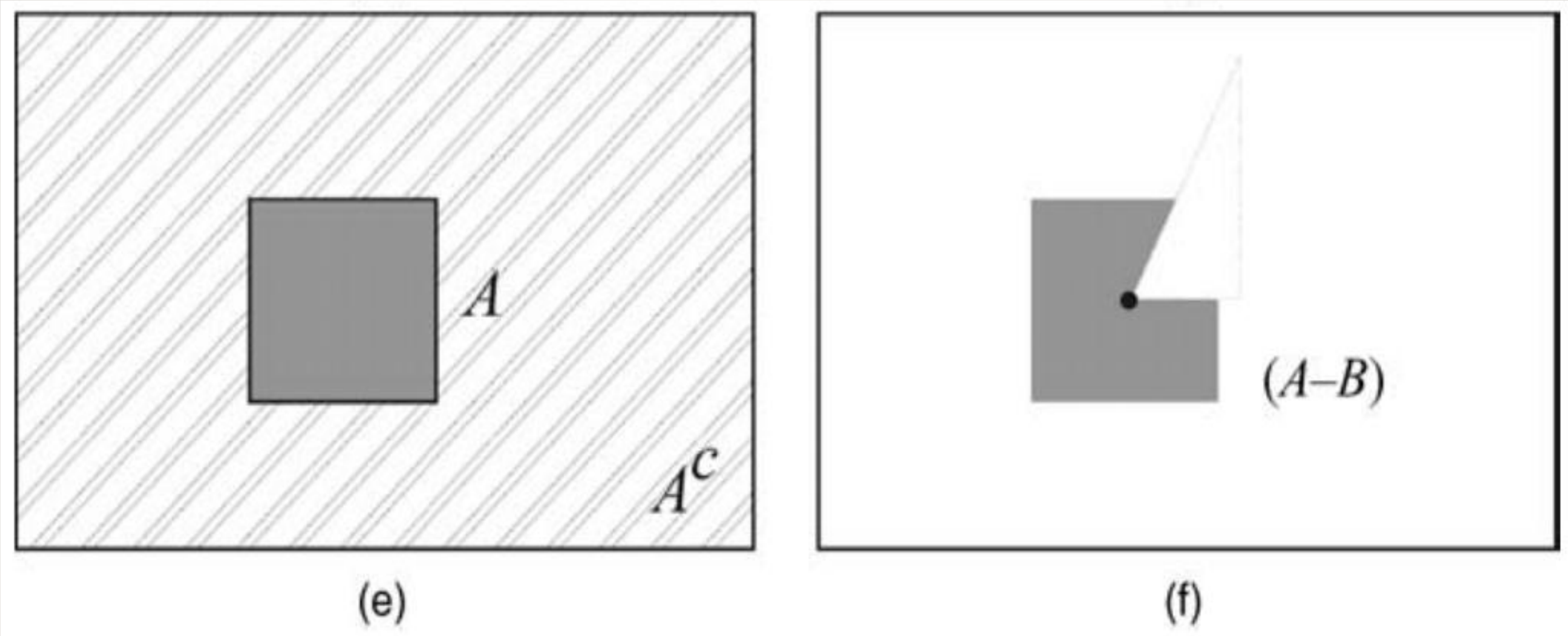


FIGURE 8.1 Basic set operations: (e) set A and its complement A^c; (f) set difference (A–B).

8.1 FUNDAMENTAL CONCEPTS AND OPERATIONS

- The statement $C = A \cap B$, from a set theory perspective, means

$$C = \{(x, y) | (x, y) \in A \; and \; (x, y) \in B\}$$

- The equivalent expression using conventional image processing notation would be

$$C(x, y) = \begin{cases} 1 & if \; A(x, y) and \; B(x, y) are \; both \; 1 \\ 0 & otherwise \end{cases}$$

- This expression leads quite easily to a single MATLAB statement that performs the *intersection* operation using the logical operator *AND* (*&*). Similarly, *complement* can be obtained using the unary *NOT* (*~*) operator, set *union* can be implemented using the logical operator *OR* (*|*), and set *difference* (*A* − *B*) can be expressed as (*A & ~B*).

8.1 FUNDAMENTAL CONCEPTS AND OPERATIONS

- Figure 8.2 shows representative results for two binary input images. Note that we have followed the IPT convention, representing foreground (1-valued) pixels as white pixels against a black background.

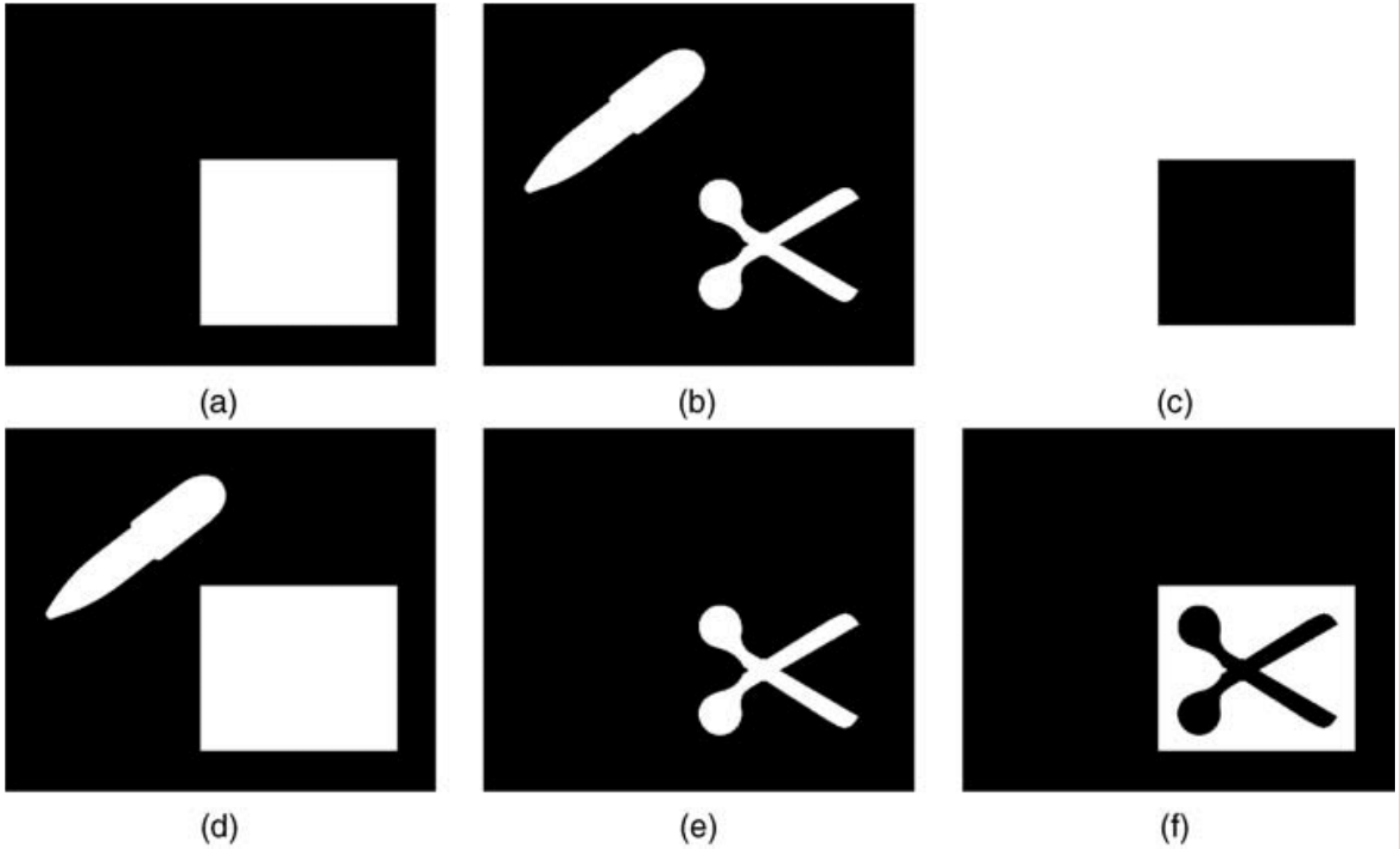


FIGURE 13.2 Logical equivalents of set theory operations: (a) Binary image (A); (b) Binary image (B); (c) Complement (A^c); (d) Union ($A \cup B$); (e) Intersection ($A \cap B$); (f) Set difference ($A - B$).

8.2.1 THE STRUCTURING ELEMENT

- The structuring element is *the basic neighborhood structure* associated with morphological image operations. It is usually represented as *a small matrix*, whose shape and size impact the results of applying a certain morphological operator to an image.
- Figure 8.3 shows two examples of SEs and how they will be represented in this lecture: the black dot corresponds to their origin (*reference point*), the gray squares represent 1 (*true*), and the white squares represent 0 (*false*).
- Although a structuring element can have any shape, its implementation requires that it should be converted to *a rectangular array*. For each array, the shaded squares correspond to the members of the SE, whereas the empty squares are used for padding, only.

8.2.1 THE STRUCTURING ELEMENT

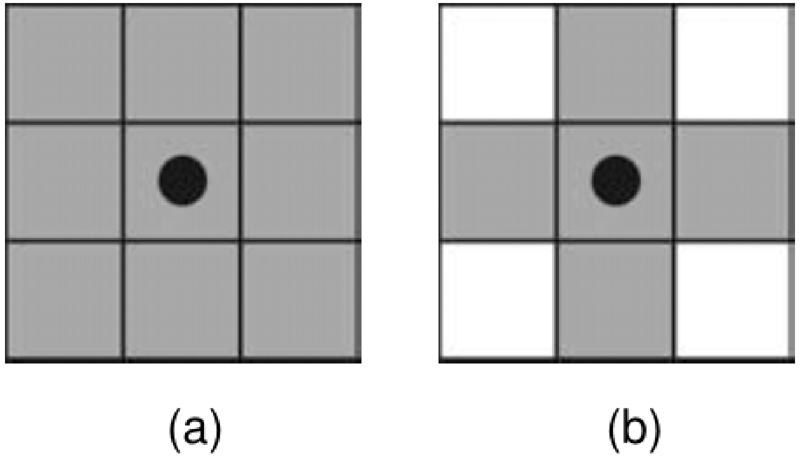


FIGURE 13.3 Examples of structuring elements: (a) square; (b) cross.

In MATLAB

MATLAB's IPT provides a function for creating structuring elements, *strel*, which supports arbitrary shapes, as well as commonly used ones, such as *square*, *diamond*, *line*, and *disk*. The result is stored as a variable of class *strel*.

EXAMPLE 8.1

- This example shows the creation of a square SE using *strel* and interpretation of its results.

```
>> se1 = strel('square',4)
se1 =
        Flat STREL object containing 16 neighbors.
        Decomposition: 2 STREL objects containing a total of 8 neighbors
        Neighborhood:
                1     1     1     1
                1     1     1     1
                1     1     1     1
                1     1     1     1
```

EXAMPLE 8.1

- The 4x4 square SE has been stored in a variable *se1*. The results displayed on the command window also indicate that the STREL object contains 16 neighbors, which can be decomposed (for faster execution) into 2 STREL objects of 8 elements each. We can then use the *getsequence* function to inspect the decomposed structuring elements.

```
>> decomp = getsequence(se1)
        decomp =
                2x1 array of STREL objects
>> decomp(1)
        ans =
                Flat STREL object containing 4 neighbors.
```

EXAMPLE 8.1

```
     Neighborhood:
          1
          1
          1
          1
>> decomp(2)
     ans =
     Flat STREL object containing 4 neighbors.
     Neighborhood:
          1     1     1     1
```

EXAMPLE 8.2

- This example shows the creation of two additional SEs of different shape and size using *strel*. Note that the rectangular SE does not require decomposition.

>> se2 = strel('diamond',4)

se2 =

Flat STREL object containing 41 neighbors.

Decomposition: 3 STREL objects containing a total of 13 neighbors

Neighborhood:

0	0	0	0	1	0	0	0	0
0	0	0	1	1	1	0	0	0
0	0	1	1	1	1	1	0	0
0	1	1	1	1	1	1	1	0
1	1	1	1	1	1	1	1	1
0	1	1	1	1	1	1	1	0
0	0	1	1	1	1	1	0	0
0	0	0	1	1	1	0	0	0
0	0	0	0	1	0	0	0	0

EXAMPLE 8.2

```
>> se3 = strel('rectangle',[1 3])
se3 =
      Flat STREL object containing 3 neighbors.
      Neighborhood:
           1     1     1
```

8.3 DILATION AND EROSION

- In this section, we discuss the two fundamental morphological operations upon which all other operations and algorithms are built: dilation and erosion.

8.3.1 DILATION

- Dilation is a morphological operation whose effect is to "grow" or "thicken" objects in a binary image. *The extent and direction of this thickening* are controlled by the size and shape of the structuring element.
- Mathematically, the dilation of a set A by B, denoted $A \oplus B$, is defined as

$$A \oplus B = \left\{ z \middle| \left(\hat{B}\right)_z \cap A \neq \emptyset \right\}$$

Figure 8.4 illustrates how dilation works using the same input object and three different structuring elements: rectangular SEs cause greater dilation along their longer dimension, as expected.

8.3.1 DILATION

In MATLAB

Morphological dilation is implemented by function *imdilate*, which takes two parameters: *an image and a structuring element.*

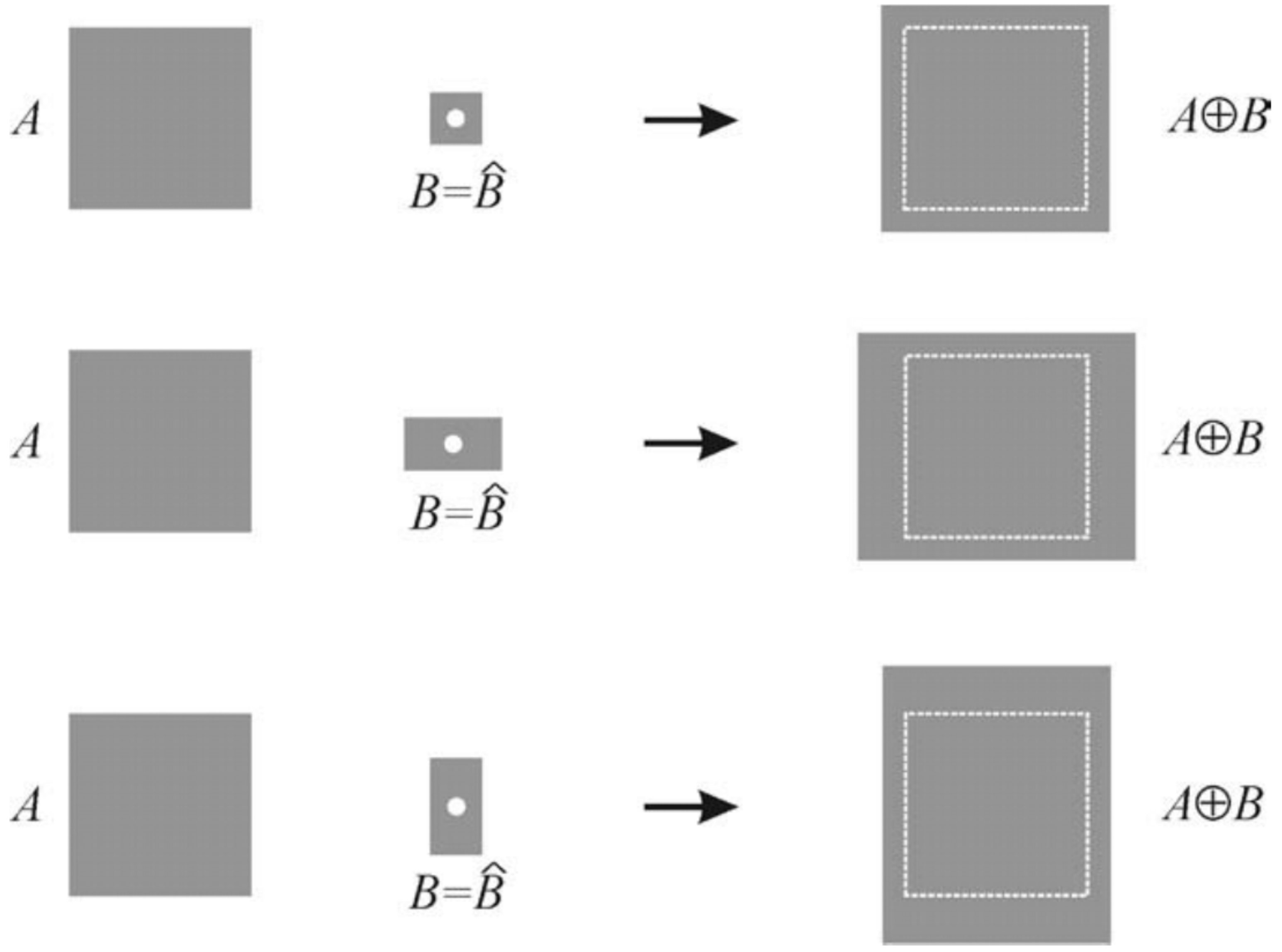


FIGURE 13.4 Example of dilation using three different rectangular structuring elements.

EXAMPLE 8.3

- This example shows the application of *imdilate* to a small binary test image with two different SEs.

```
>> a = [0 0 0 0 0;0 1 1 0 0;0 1 1 0 0;0 0 1 0 0;0 0 0 0 0]
   a=
        0    0    0    0    0
        0    1    1    0    0
        0    1    1    0    0
        0    0    1    0    0
        0    0    0    0    0
```

EXAMPLE 8.3

```
>> se1 = strel('square',2)
se1 =
        Flat STREL object containing 4 neighbors.
        Neighborhood:
                1    1
                1    1
>> b = imdilate(a,se1)
        b =
                0    0    0    0    0
                0    1    1    1    0
                0    1    1    1    0
                0    1    1    1    0
                0    0    1    1    0
```

EXAMPLE 8.3

- The dilation of a with SE *se1* (whose reference pixel is the top left corner) caused every pixel in a whose value was 1 to be replaced with a 2 x 2 square of pixels equal to 1.

```
>> se2 = strel('rectangle', [1 2])
    se2 =
        Flat STREL object containing 2 neighbors.
        Neighborhood:
            1    1
```

EXAMPLE 8.3

>> c = imdilate(a,se2)

c =

0	0	0	0	0
0	1	1	1	0
0	1	1	1	0
0	0	1	1	0
0	0	0	0	0

- The dilation of a with SE *se2* (whose reference pixel is the leftmost one) caused every pixel in a whose value was 1 to be replaced with a 1 x 2 rectangle of pixels equal to 1, that is, *a predominantly horizontal dilation*.

8.3.2 EROSION

- Erosion is a morphological operation whose effect is to "shrink" or "thin" objects in a binary image. *The direction and extent of this thinning* is controlled by the shape and size of the structuring element.
- Mathematically, the erosion of a set A by B, denoted $A \ominus B$, is defined as

$$A \ominus B = \left\{ z \middle| \left(\hat{B}\right)_z \cup A^c \neq \emptyset \right\}$$

Figure 8.5 illustrates how erosion works using the same input object and three different structuring elements. Erosion is more severe in the direction of the longer dimension of the rectangular SE.

8.3.2 EROSION

In MATLAB

Morphological erosion is implemented by function *imerode*, which takes two parameters: *an image and a structuring element.*

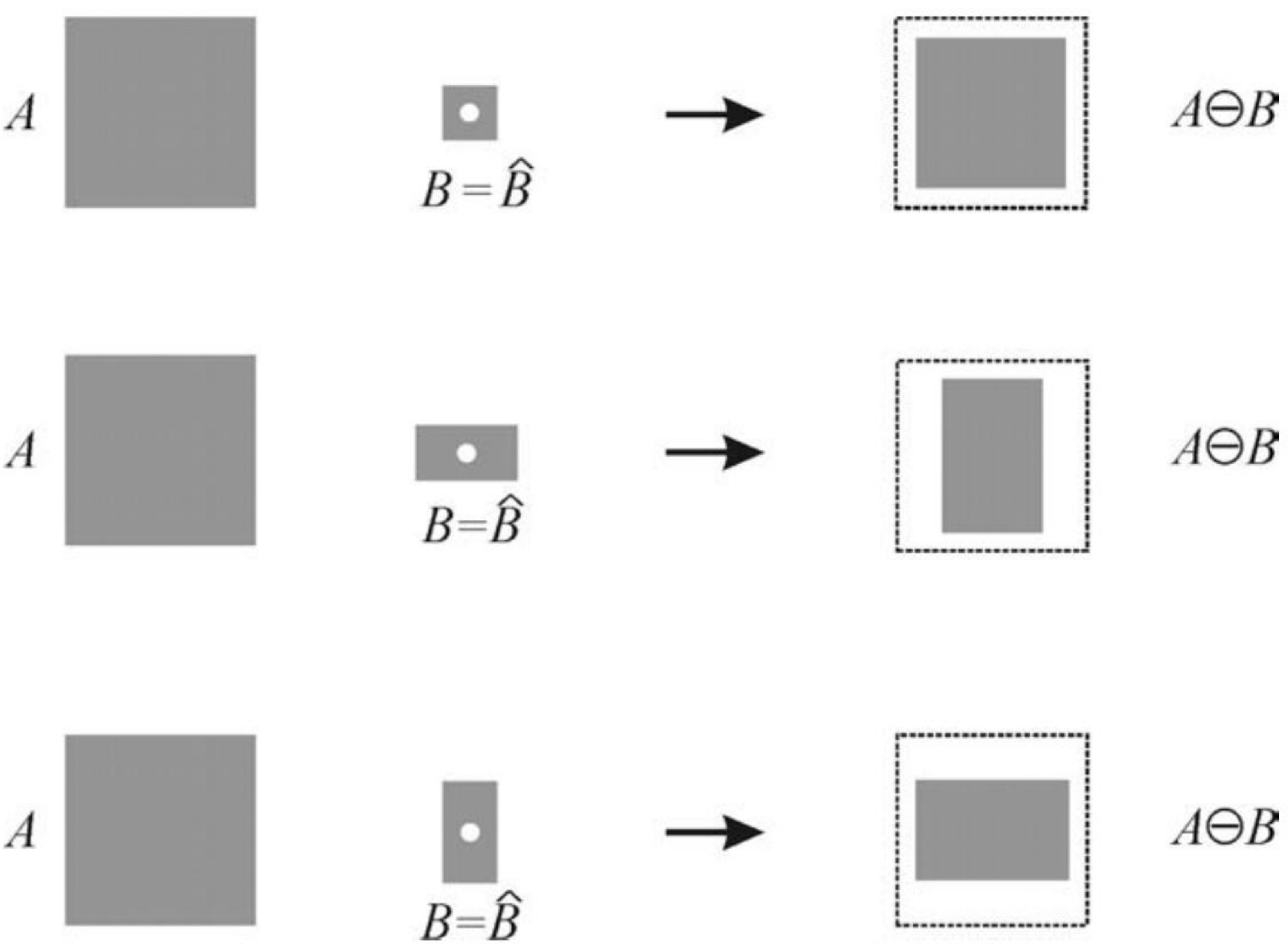


FIGURE 13.5 Example of erosion using three different rectangular structuring elements.

EXAMPLE 8.4

This example shows the application of *imerode* to a small binary test image with two different SEs.

```
>> a = [0 0 0 0 0;0 1 1 1 0;1 1 1 0 0;0 1 1 1 1;0 0 0 0 0]
   a =
         0   0   0   0   0
         0   1   1   1   0
         1   1   1   0   0
         0   1   1   1   1
         0   0   0   0   0
```

EXAMPLE 8.4

>> se1 = strel('square',2)
se1 =
Flat STREL object containing 4 neighbors.
Neighborhood:

1	1
1	1

>> b = imerode(a,se1)
b =

0	0	0	0	0
0	1	0	0	0
0	1	0	0	0
0	0	0	0	0
0	0	0	0	0

EXAMPLE 8.4

- The erosion of a with SE *se1* (whose reference pixel is the top left corner) caused the disappearance of many pixels in a; the ones that remained were the ones corresponding to the top left corner of a 2 x 2 square of pixels equal to 1.

>> se2 = strel('rectangle', [1 2])

se2 =

Flat STREL object containing 2 neighbors.

Neighborhood:

1 1

EXAMPLE 8.4

>> c = imerode(a,se2)

c =

0	0	0	0	0
0	1	1	0	0
1	1	0	0	0
0	1	1	1	1
0	0	0	0	0

- The erosion of a with SE *se2* (whose reference pixel is the leftmost one) caused the disappearance of many pixels in a; the ones that remained were the ones corresponding to the leftmost pixel of a 1x2 rectangle of pixels equal to 1.

8.3.2 EROSION

- Erosion is the dual operation of dilation and vice versa:

$$(A \ominus B)^c = A^c \oplus \hat{B}$$

$$A \oplus B = \left(A^c \ominus \hat{B}\right)^c$$

- Erosion and dilation can also be interpreted in terms of whether a SE *hits* or *fits* an image (region).

8.3.2 EROSION

- For dilation, the resulting image $g(x,y)$, given an input image $f(x,y)$ and a SE *se*, will be

$$g(x,y)=\begin{cases}1 \; if \; se \; hits \; f \\ 0 \;\; otherwise\end{cases}$$

for all x and y.

- For erosion, the resulting image $g(x,y)$, given an input image $f(x,y)$ and a SE *se*, will be

$$g(x,y)=\begin{cases}1 \; if \; se \; fits \; f \\ 0 \;\; otherwise\end{cases}$$

for all x and y.

8.4 COMPOUND OPERATIONS

- In this section, we present morphological operations that combine the two fundamental operations (erosion and dilation) in different ways.

8.4.1 OPENING

- The morphological opening of set A by B, represented as $A \circ B$, is *the erosion of A by B followed by the dilation of the result by B.* Mathematically,

$$A \circ B = (A \ominus B) \oplus B$$

- Alternatively, the opening operation can be expressed using set notation as

$$A \circ B = \bigcup \{(B)_z \mid (B)_z \subseteq A\}$$

where $\cup\{\cdot\}$ represents the union of all sets within the curly braces, and the symbol $\subseteq$ means "*is a subset of.*"

8.4.1 OPENING

- The opening operation is *idempotent*, that is, once an image has been opened with a certain SE, subsequent applications of the opening algorithm with the same SE will not cause any effect on the image. Mathematically,

$$(A \circ B) \circ B = A \circ B$$

- Morphological opening is typically used to remove thin protrusions from objects and to open up a gap between objects connected by a thin bridge without shrinking the objects (as erosion would have done). It also causes a smoothening of the object's contour (Figure 13.6).
- The geometric interpretation of the opening operation is straightforward: $A \circ B$ is the union of all translations of B that fit entirely within A.

8.4.1 OPENING

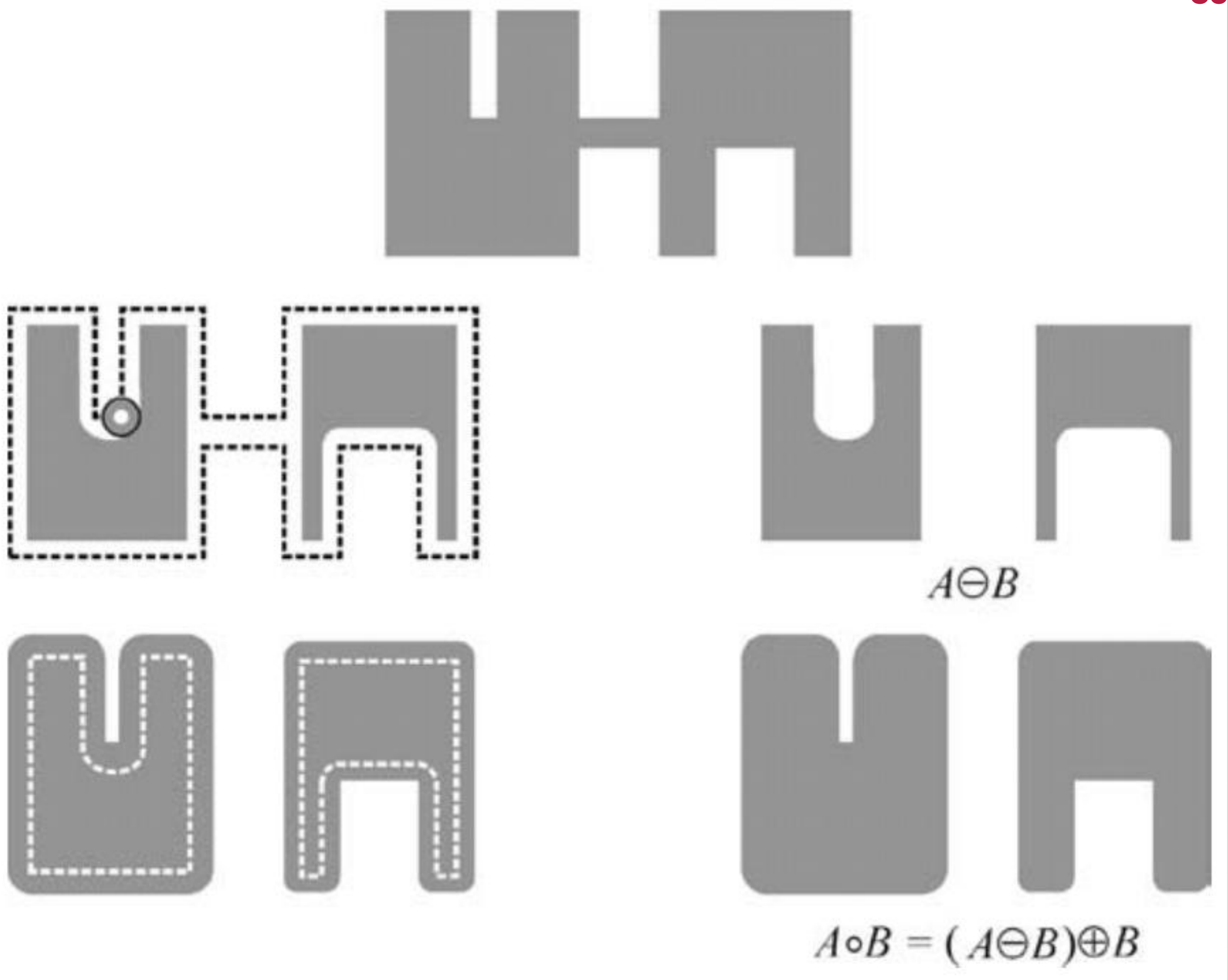


FIGURE 13.6 Example of morphological opening.

8.4.1 OPENING

In MATLAB

Morphological opening is implemented by function *imopen*, which takes two parameters: *input image and structuring element.*

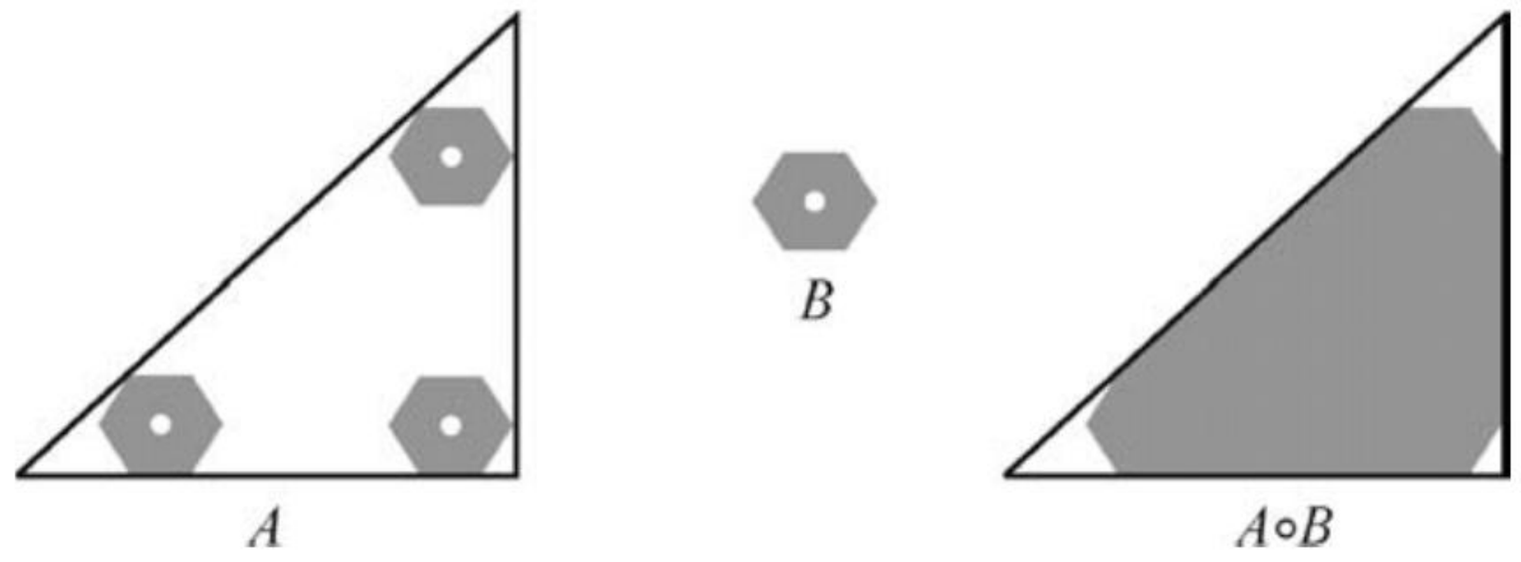


FIGURE 13.7 Geometric interpretation of the morphological opening operation.

8.4.2 CLOSING

- The morphological closing of set A by B, represented as $A \bullet B$, is *the dilation of* A *by* B *followed by the erosion of the result by* B. Mathematically,

$$A \bullet B = (A \oplus B) \ominus B$$

- Just as with opening, the closing operation is *idempotent*, that is, once an image has been closed with a certain SE, subsequent applications of the opening algorithm with the same SE will not cause any effect on the image. Mathematically,

$$(A \bullet B) \bullet B = A \bullet B$$

8.4.2 CLOSING

- Morphological closing is typically used to fill small *holes*, fuse narrow breaks, and close thin gaps in the objects within an image, without changing the objects' size (as dilation would have done). It also causes a smoothening of the object's contour.
- The geometric interpretation of the closing operation is as follows: $A \bullet B$ is the complement of the union of all translations of B that do not overlap A.
- Closing is the dual operation of opening and vice versa:

$$A \bullet B = (A^c \circ B)^c$$

$$A \circ B = (A^c \bullet B)^c$$

8.4.2 CLOSING

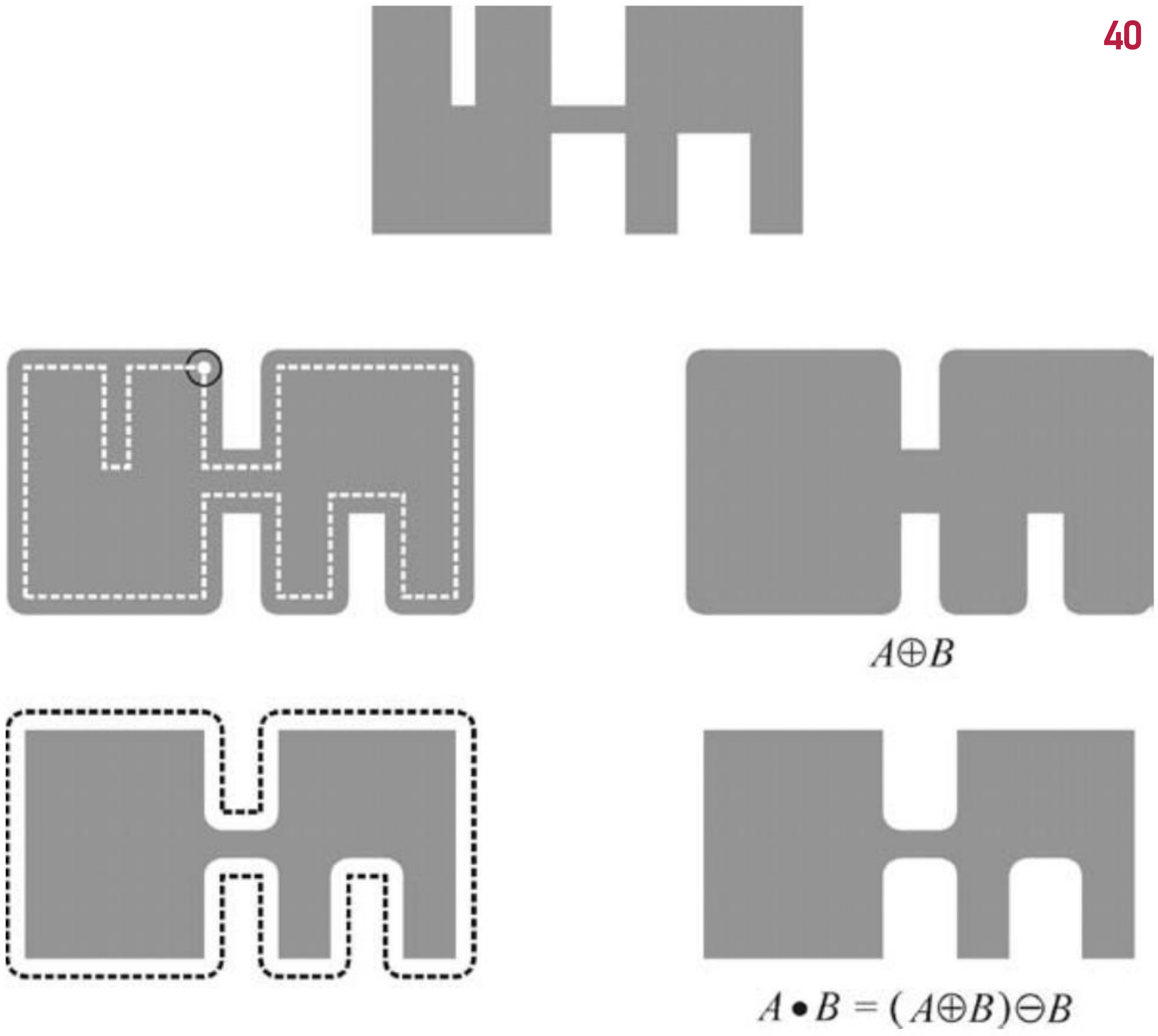


FIGURE 13.8 Example of morphological closing.

8.4.2 CLOSING

In MATLAB

Morphological closing is implemented by function *imclose*, which takes two parameters: *input image and structuring element.*

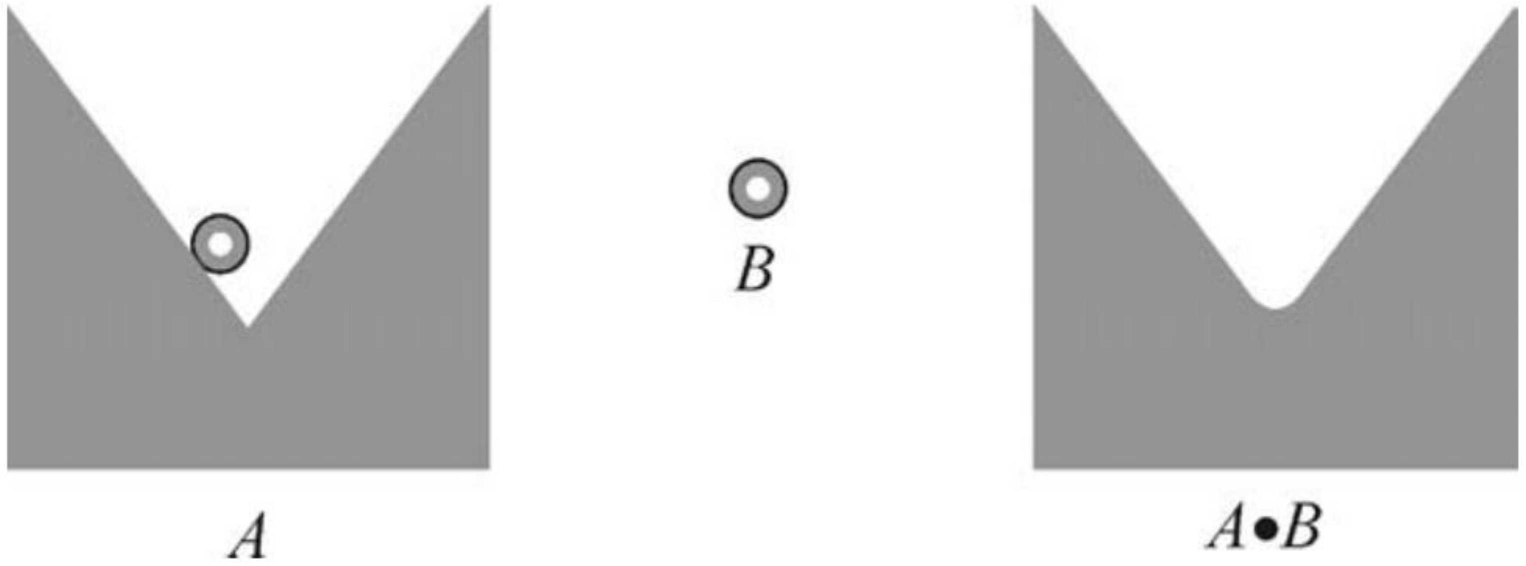


FIGURE 13.9 Geometric interpretation of the morphological closing operation. Adapted and redrawn from [GW08].

8.4.3 HIT-OR-MISS TRANSFORM

- The hit-or-miss (HoM) transform is a combination of morphological operations that uses two structuring elements (B_1 and B_2) designed in such a way that the output image will consist of all locations that match the pixels in B_1 (a hit) and that have none of the pixels in B_2 (a miss).
- Mathematically, the HoM transform of image A by the structuring element set B ($B = (B_1, B_2)$), denoted $A \otimes B$, is defined as

$$A \otimes B = (A \ominus B_1) \cap (A^c \ominus B_2)$$

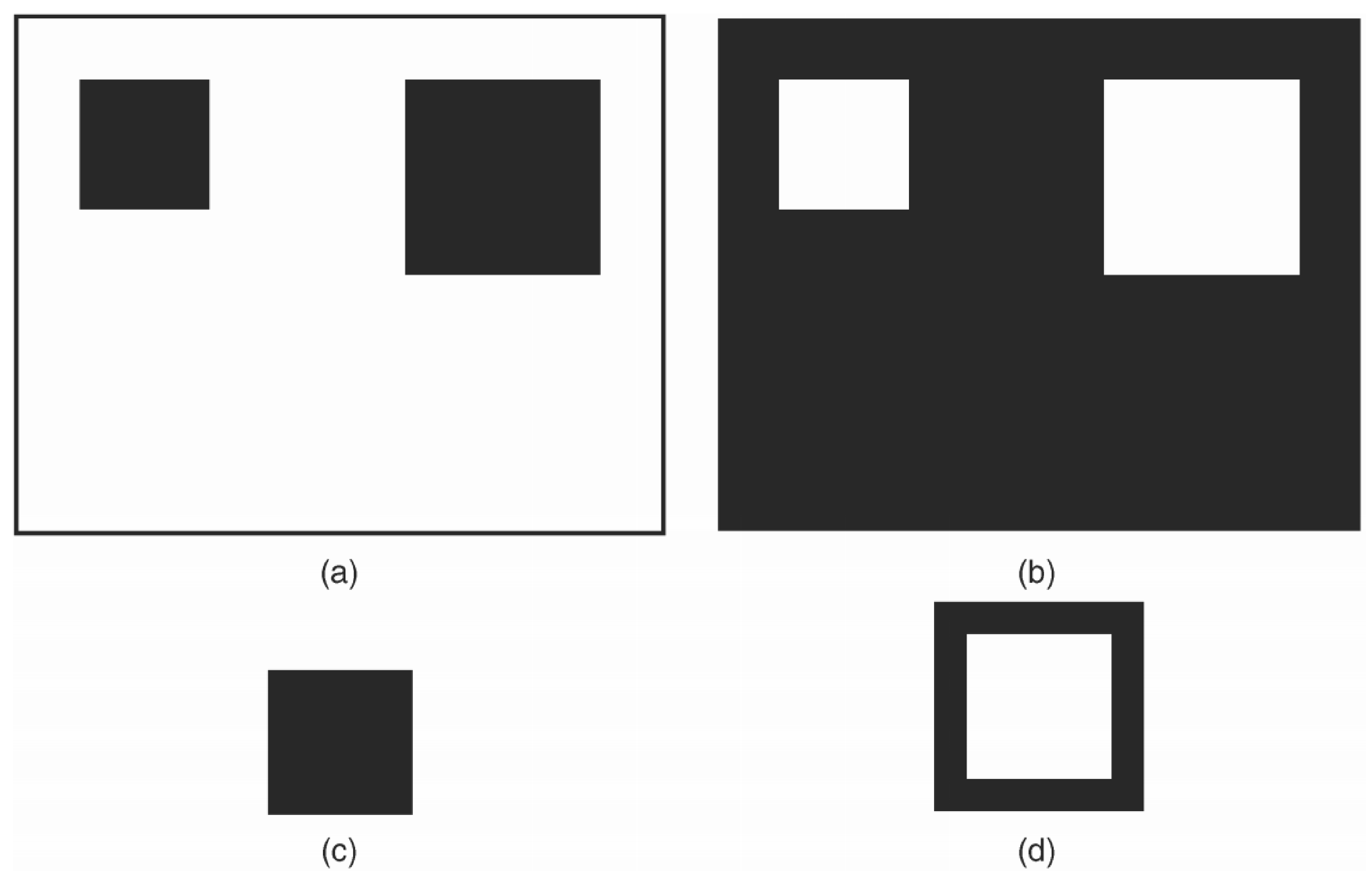
- Alternatively, the HoM transform can be expressed as

$$A \otimes B = (A \ominus B_1) - \left(A \oplus \hat{B}_2\right)$$

8.4.3 HIT-OR-MISS TRANSFORM

- Figure 8.10 shows how the HoM transform can be used to locate squares of a certain size in a binary image.
- Figure 8.10a shows the input image (A), consisting of two squares against a background, and Figure 8.10b shows its complement (A^c).
- The two structuring elements, in this case, have been chosen to be a square of the same size as the smallest square in the input image, B_1 (Figure 8.10c), and a square of the larger size, indicating that the smallest square should be surrounded by pixels of opposite color, B_2 (Figure 8.10d).

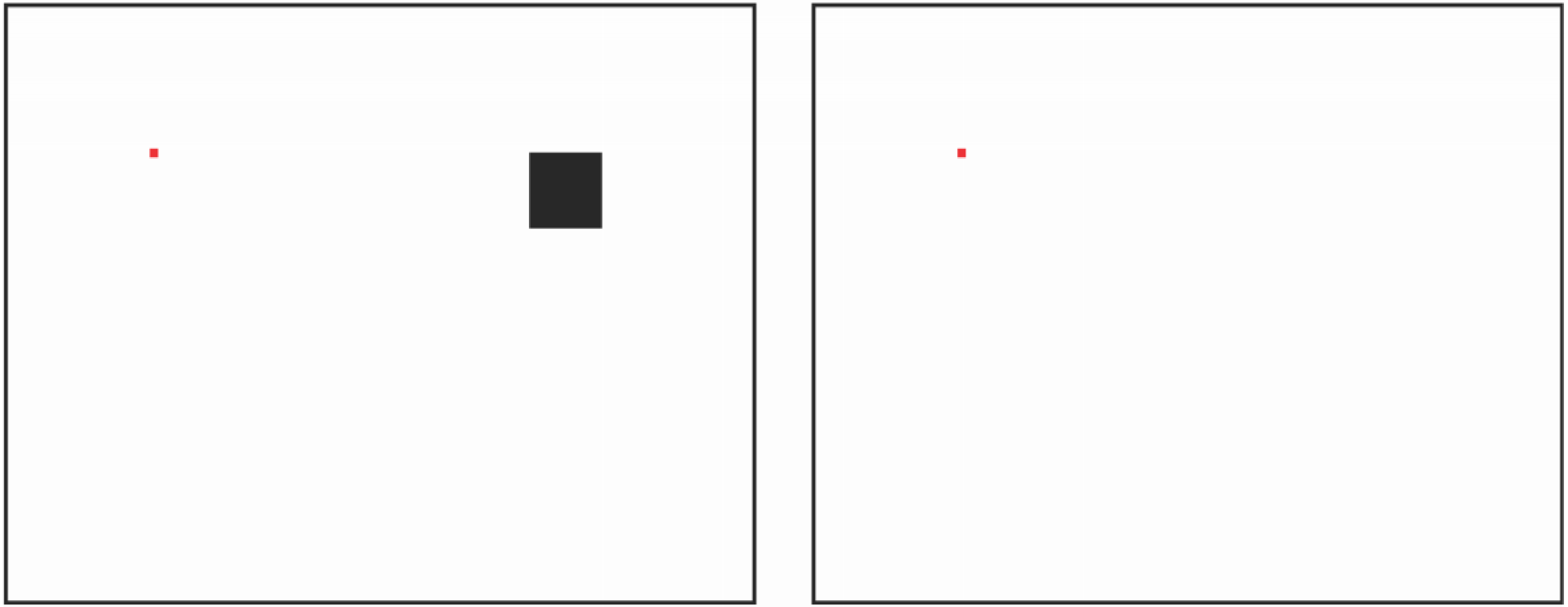
8.4.3 HIT-OR-MISS TRANSFORM



8.4.3 HIT-OR-MISS TRANSFORM

- Figure 8.10e shows the partial result after the first erosion: the smallest square is hit at a single spot, magnified, and painted red for viewing purposes, whereas the largest square is hit at many points.
- Figure 8.10f shows the final result, containing the set of all points for which the HoM transform found a hit for B_1 in A and a hit for B_2 in A^c (which is equivalent to a miss for B_2 in A).

8.4.3 HIT-OR-MISS TRANSFORM



(e) (f)

FIGURE 13.10 Example of HoM transform.

8.4.3 HIT-OR-MISS TRANSFORM

In MATLAB

- The binary hit-or-miss transformation is implemented by function *bwhitmiss*, whose basic syntax is *J = bwhitmiss(I, B1, B2)*, where I is the input image, $B1$ and $B2$ are the structuring elements, and J is the resulting image.

8.5 MORPHOLOGICAL FILTERING

- Morphological filters are Boolean filters that apply a many-to-one binary (or Boolean) function h within a window W in the binary input image $f(x, y)$, producing at the output an image $g(x, y)$ given by

$$g(x, y) = h[Wf(x, y)]$$

- Examples of Boolean operations (denoted as h in above equation) are as follows:
  - OR: This is equivalent to a morphological dilation with a square SE of the same size as W.
  - AND: This is equivalent to a morphological erosion with a square SE of the same size as W.
  - MAJ (Majority): This is the morphological equivalent to a median filter applicable to binary images.

8.5 MORPHOLOGICAL FILTERING

- Morphological filters can also be used in noise reduction. Let A be a binary image corrupted by impulse (salt and pepper) noise. The application of a morphological opening operation followed by a morphological closing will remove a significant amount of noise present in the input image, resulting in an image C given by

$$C = \big((A \circ B) \bullet B\big)$$

where B is the chosen SE.

EXAMPLE 8.5

- Figure 8.11 shows an example of morphological filtering for noise removal on a binary image using a circular SE of radius equal to 2 pixels.
- Part (a) shows the (641 x 535) input image, contaminated by salt and pepper noise. Part (b) shows the partial result (after the opening operation): at this stage, all the "salt" portion of the noise has been removed, but the "pepper" noise remains.
- Finally, part (c) shows the result of applying closing to the result of the opening operation. The noise has been completely removed, at the expense of imperfections at the edges of the objects. Such imperfections would be even more severe for larger SEs, as demonstrated in part (d), where the radius of the circular SE is equal to 4 pixels.

EXAMPLE 8.5

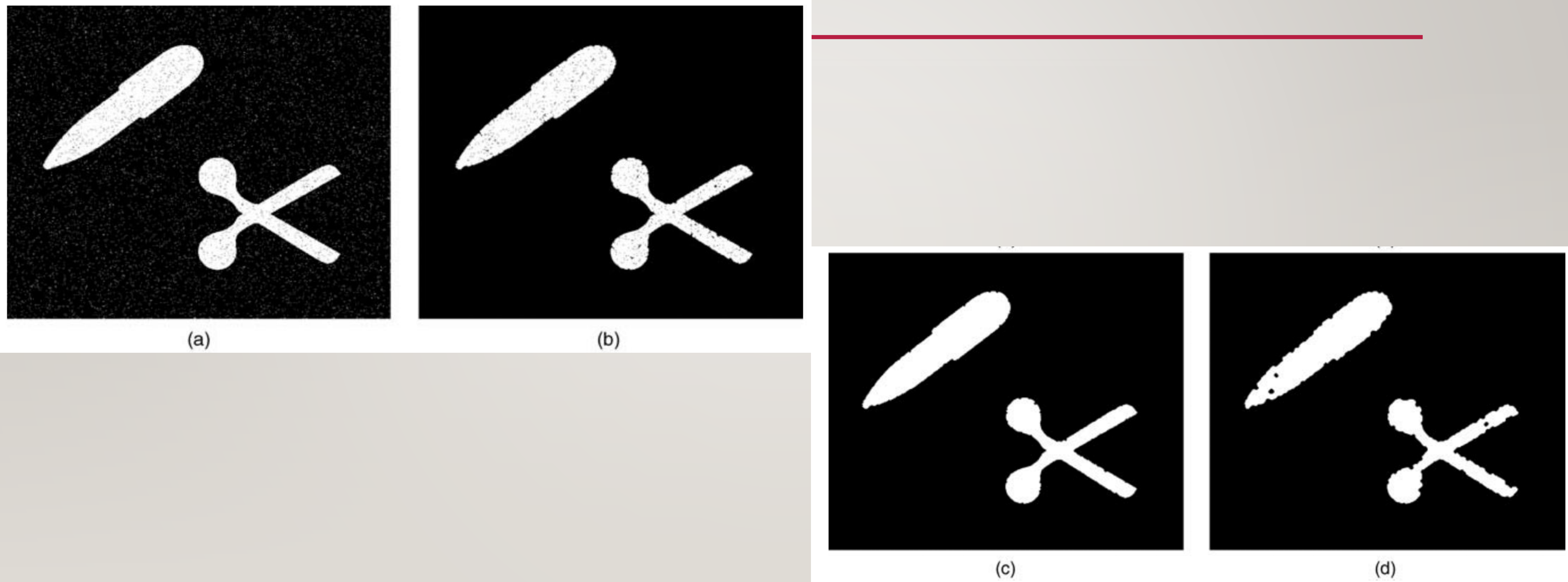


FIGURE 13.11 Morphological filtering. (a) input (noisy) image; (b) partial result (after opening) with SE of radius = 2 pixels; (c) final result with SE of radius = 2 pixels; (d) final result with SE of radius = 4 pixels.

8.6 BASIC MORPHOLOGICAL ALGORITHMS

- In this section, we present a collection of simple and useful morphological algorithms and show how they can be implemented using MATLAB and the IPT.

In MATLAB

- The IPT function *bwmorph* implements a number of useful morphological operations and algorithms, listed in Table 8.1. This function takes three arguments: input image, desired operation, and the number of times the operation is to be repeated.

8.6 BASIC MORPHOLOGICAL ALGORITHMS

TABLE 13.1 Operations Supported by bwmorph

Operation	Description
`bothat`	Subtract the input image from its closing
`bridge`	Bridge previously unconnected pixels
`clean`	Remove isolated pixels (1s surrounded by 0s)
`close`	Perform binary closure (dilation followed by erosion)
`diag`	Diagonal fill to eliminate 8-connectivity of background
`dilate`	Perform dilation using the structuring element 1s(3)
`erode`	Perform erosion using the structuring element 1s(3)
`fill`	Fill isolated interior pixels (0s surrounded by 1s)
`hbreak`	Remove H-connected pixels
`majority`	Set a pixel to 1 if five or more pixels in its 3×3 neighborhood are 1s
`open`	Perform binary opening (erosion followed by dilation)
`remove`	Set a pixel to 0 if its 4-connected neighbors are all 1s, thus leaving only boundary pixels

8.6 BASIC MORPHOLOGICAL ALGORITHMS

`shrink`	With N = Inf, shrink objects to points; shrink objects with holes to connected rings
`skel`	With N = Inf, remove pixels on the boundaries of objects without allowing objects to break apart
`spur`	Remove endpoints of lines without removing small objects completely
`thicken`	With N = Inf, thicken objects by adding pixels to the exterior of objects without connecting previously unconnected objects
`thin`	With N = Inf, remove pixels so that an object without holes shrinks to a minimally connected stroke, and an object with holes shrinks to a ring halfway between the hold and outer boundary
`tophat`	Subtract the opening from the input image

EXAMPLE 8.6

- In this example, we use *bwmorph* to apply different operations to the same small test image (Figure 8.12a).
- Here is the sequence of steps used to obtain the results in Figure 8.12(b–f):

B = bwmorph(A,'skel', Inf);

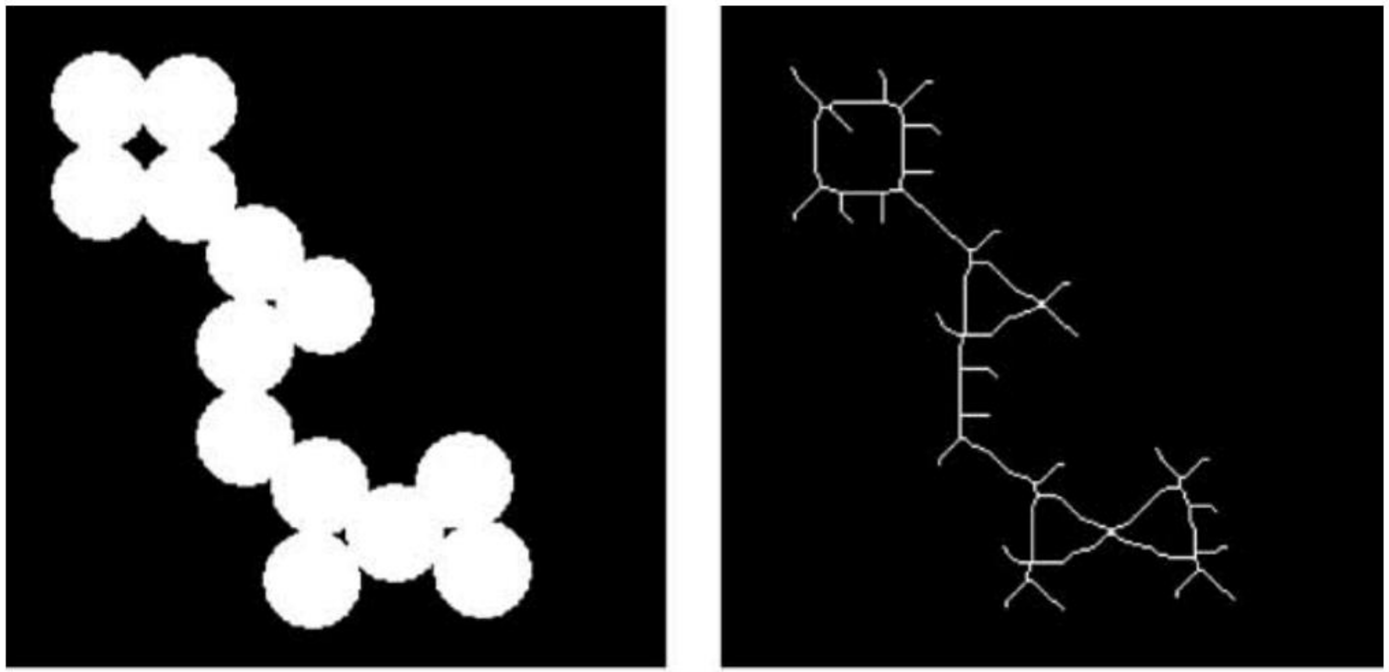
C = bwmorph(B,'spur',Inf);

D = bwmorph(A,'remove');

E = bwmorph(D,'thicken',3);

F = bwmorph(E,'thin',3);

EXAMPLE 8.6



(a) (b)

FIGURE 8.12 Morphological algorithms. (a) input image; (b) skeleton of (a);

EXAMPLE 8.6

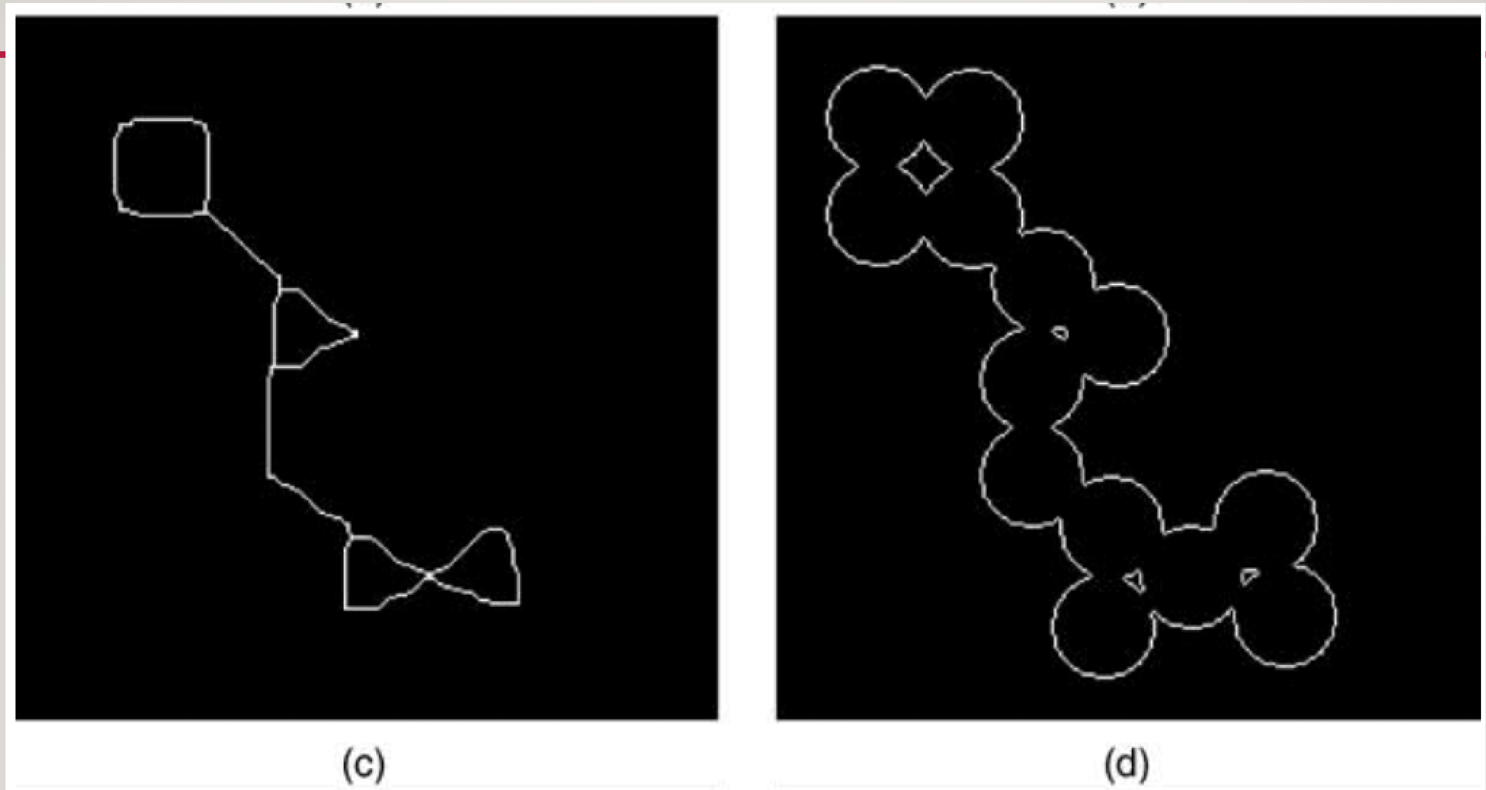


FIGURE 8.12 Morphological algorithms. (c) pruning spurious pixels from (b); (d) removing interior pixels from (a);

EXAMPLE 8.6

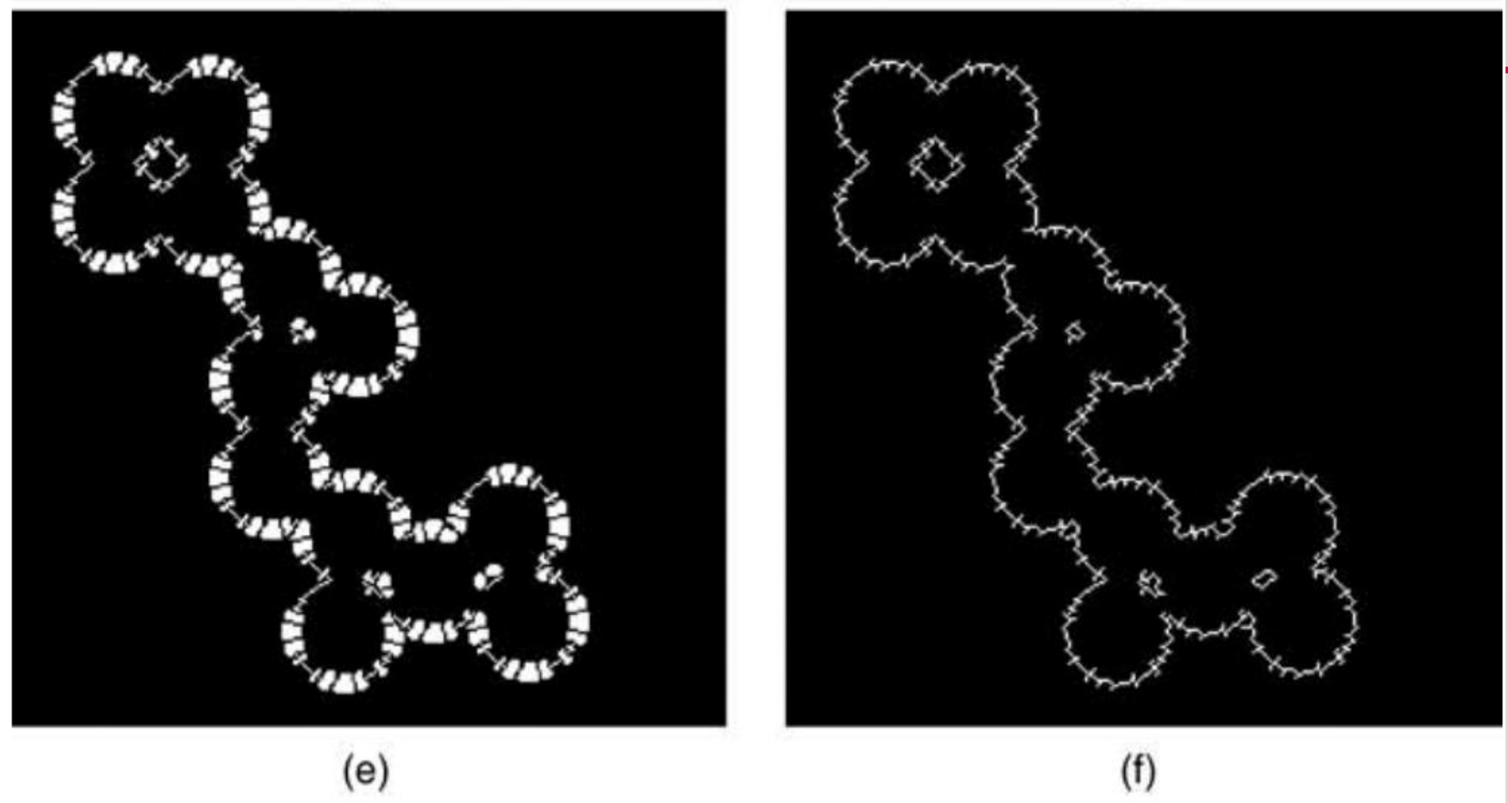


FIGURE 8.12 Morphological algorithms. (e) thickening the image in (d); (f) thinning the image in (e). Original image: courtesy of MathWorks.

8.6.1 BOUNDARY EXTRACTION

- Dilation, erosion, and set difference operations can be combined to perform boundary extraction of a set A, denoted by $\mathcal{BE}(A)$, as follows:
- *Internal Boundary*: Consists of the pixels in A that sit at the edge of A.

$$\mathcal{BE}(A) = A - (A \ominus B)$$

- *External Boundary*: Consists of the pixels outside A that sit immediately next to A.

$$\mathcal{BE}(A) = (A \oplus B) - A$$

- *Morphological Gradient*: Consists of the combination of internal and external boundaries.

$$\mathcal{BE}(A) = (A \oplus B) - (A \ominus B)$$

where B is a suitable structuring element.

8.6.1 BOUNDARY EXTRACTION

- Figure 8.13 shows an example of internal boundary extraction of a small (5 x 12) object using a 3 x 3 square as SE.

In MATLAB

- The IPT function *bwperim* returns a binary image containing only the perimeter pixels of objects in the input image.

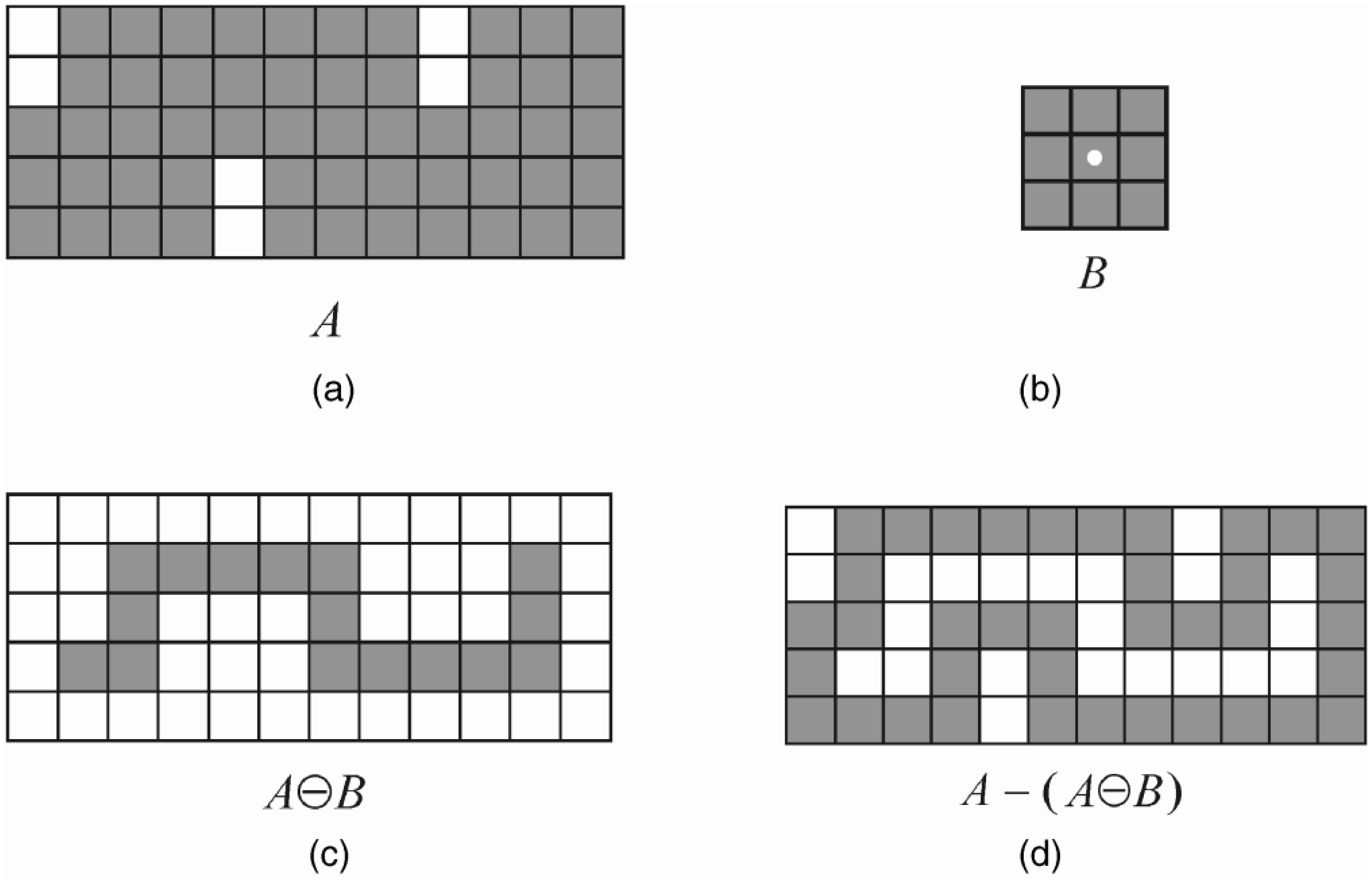


FIGURE 13.13 Boundary extraction.

EXAMPLE 8.7

- This example shows the application of *bwperim* (with 8-connectivity, to be consistent with the SE in Figure 8.13b) to extract the internal boundary of a small binary test image identical to the image in Figure 8.13a.

```
a = ones(5,12)
a (1:2,1)=0
a (1:2,9)=0
a (4:5,5)=0
b = bwperim(a,8)
```

- When you execute the steps above, you will confirm that the result (stored in variable b) is identical to the image in Figure 8.13d.

8.6.2 REGION FILLING

- In this section, we present an algorithm that uses morphological and set operations to fill regions (which can be thought of as *holes*) in a binary image.
- Let p be a pixel in a region surrounded by an 8-connected boundary, A. The goal of a region filling algorithm is to fill up the entire region with 1s using p as a starting point (i.e., setting it as 1). Region filling can be accomplished using an iterative procedure, mathematically expressed as follows:

$$X_k = (X_{k-1} \oplus B) \cap A^c \qquad k = 1,2,3,\ldots$$

where $X_0 = p$ and B is the cross-shaped structuring element. The algorithm stops at the kth iteration if $X_k = X_{k-1}$. The union of X_k and A contains the original boundary (A) and all the pixels within it labeled as 1.

8.6.2 REGION FILLING

- Figure 8.14 illustrates the process. Part (a) shows the input image, whose complement is in part (b).
- Part (c) shows the partial results of the algorithm after each iteration (the number inside the square): the initial pixel (p, top left) corresponds to iteration 0. This example requires six iterations to complete and uses the SE shown in part (e). Part (d) shows the union of X_6 and A.

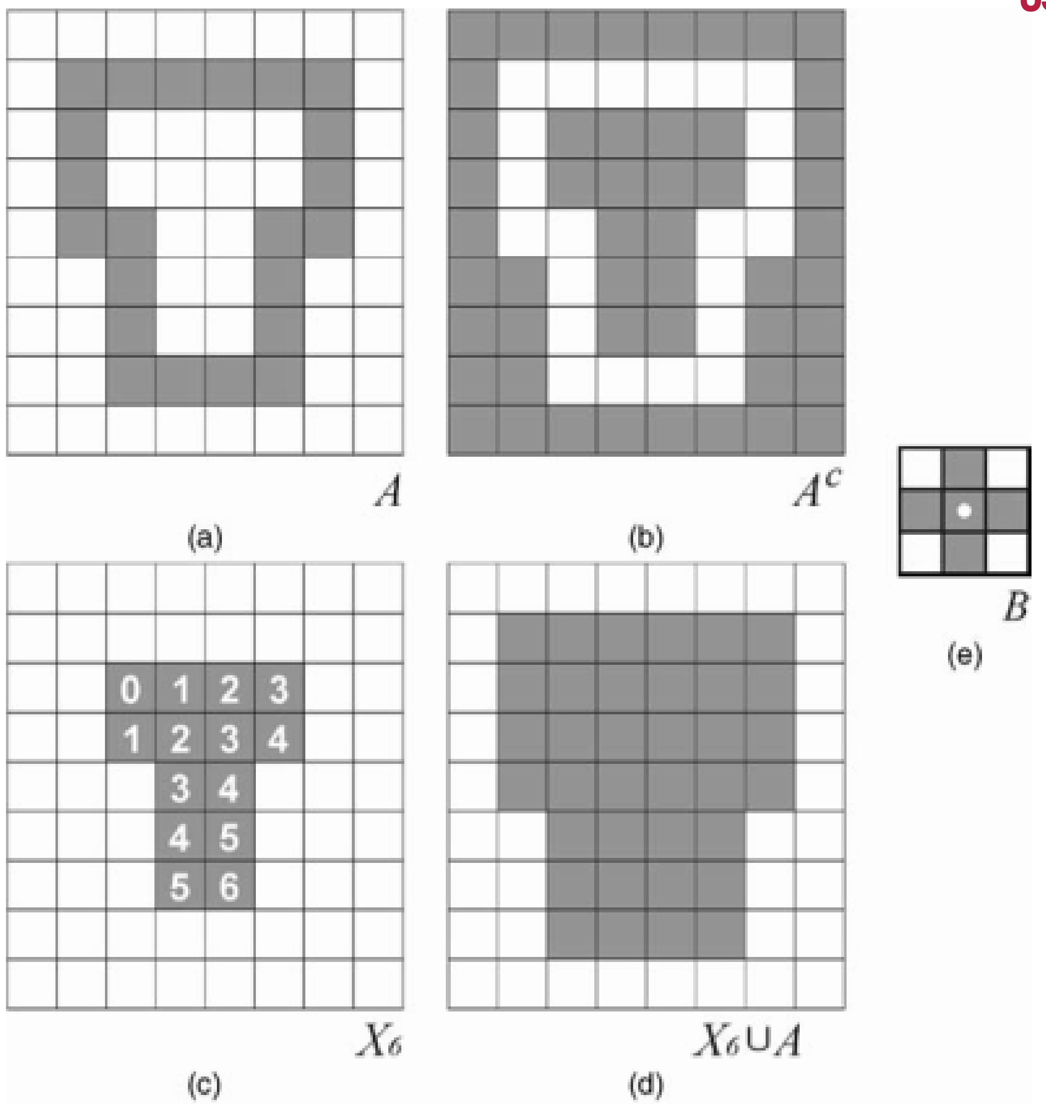


FIGURE 13.14 Region filling: (a) input image; (b) complement of (a); (c) partial results (numbered according to the iteration in the algorithm described by equation (13.27); (d) final result; (e) structuring element.

8.6.2 REGION FILLING

In MATLAB

- The IPT function *imfill* implements region filling. It can be used in an interactive mode (where the user clicks on the image pixels that should be used as starting points) or by passing the coordinates of the starting points.

8.6.3 EXTRACTION AND LABELING OF CONNECTED COMPONENTS

- Morphological concepts and operations can be used to extract and label connected components in a binary image.
- The morphological algorithm for extraction of connected components is very similar to the region filling algorithm described in Section 8.6.2. Let A be a set containing one or more connected components and p $(p \in A)$ be a starting pixel. The process of finding all other pixels in a component can be accomplished using an iterative procedure, mathematically expressed as follows:

$$X_k = (X_{k-1} \oplus B) \cap A \qquad k = 1,2,3, \ldots$$

where $X_0 = p$ and B is a suitable structuring element: cross-shaped for 4-connectivity, 3 x 3 square for 8-connectivity.
- The algorithm stops at the kth iteration if $X_k = X_{k-1}$.

8.6.3 EXTRACTION AND CONNECTED COMPON

- Figure 8.15 shows an example of extraction of connected components.
- Part (a) shows the initial set A and the starting pixel p, represented by the number 0. Part (e) shows the SE used in this case. Parts (b) and (c) show the result of the first and second iterations, respectively. The final result (after six iterations) is shown in Figure 8.15d.

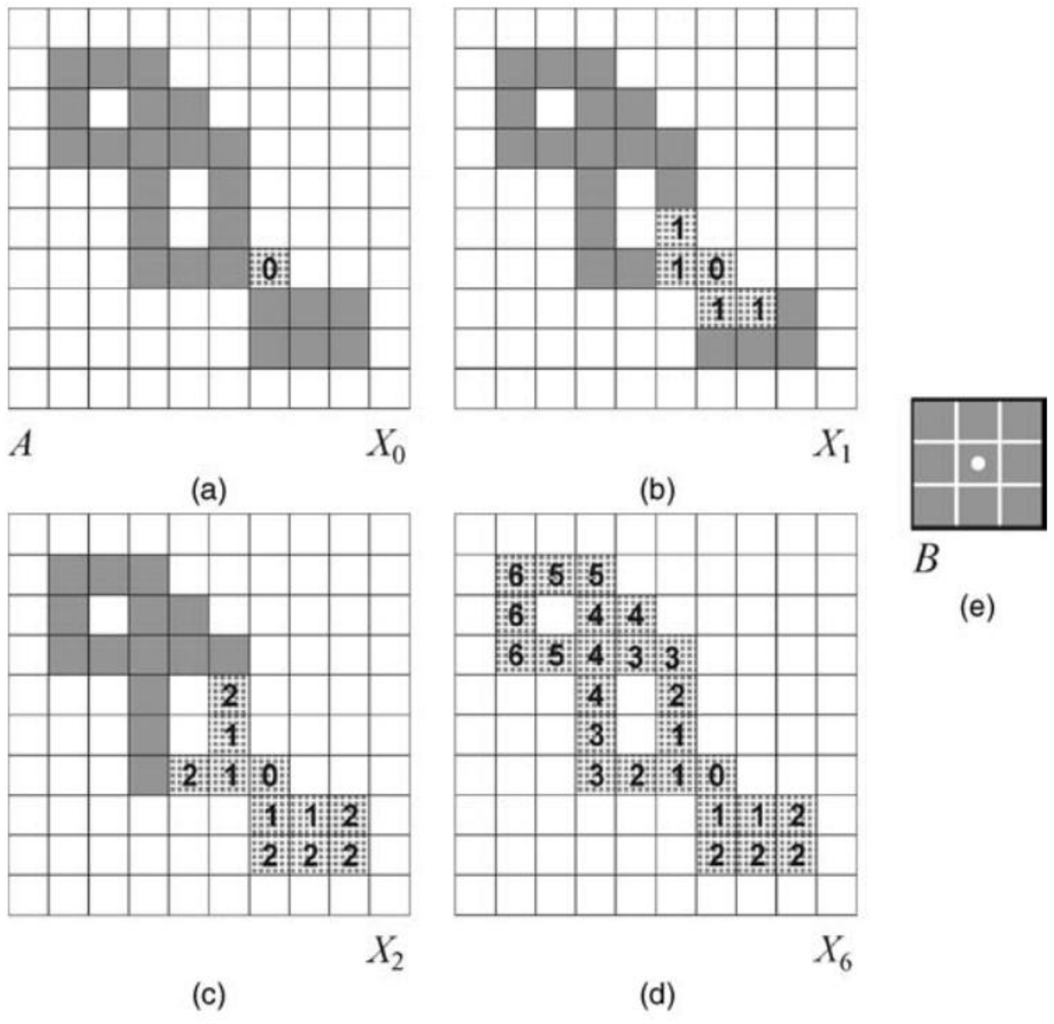


FIGURE 13.15 Extraction of connected components: (a) input image; (b) first iteration; (c) second iteration; (d) final result, showing the contribution of each iteration (indicated by the numbers inside the squares); (e) structuring element.

8.7 GRAYSCALE MORPHOLOGY

- Many morphological operations originally developed for binary images can be extended to grayscale images. This section presents the grayscale version of the basic morphological operations (erosion, dilation, opening, and closing) and introduces the top-hat and bottom-hat transformations.
- The mathematical formulation of grayscale morphology uses an input image $f(x, y)$ and a structuring element (SE) $b(x, y)$. Structuring elements in grayscale morphology come in two categories: *nonflat* and *flat*.

In MATLAB

- Nonflat SEs can be created with the same function used to create flat SEs: *strel*. In the case of nonflat SEs, we must also pass a second matrix (containing the height values) as a parameter.

8.7.1 DILATION AND EROSION

- The dilation of an image $f(x, y)$ by a flat SE $b(x, y)$ is defined as

$$[A \oplus B](x, y) = max\{f(x+s, y+t) | (s,t) \in D_b\} \quad (8.1)$$

where D_b is called the domain of b.

- For a nonflat SE, $b_N(x, y)$, equation (8.1) becomes

$$[A \oplus B](x, y) = max\{f(x+s, y+t) + b_N(s,t) | (s,t) \in D_b\} \quad (8.2)$$

- Some references use $f(x-s, y-t)$ in equations (8.1) and (8.2), which just requires the SE to be rotated by 180° or mirrored around its origin, that is, $\hat{B}(x, y) = b(-x, -y)$.

8.7.1 DILATION AND EROSION

- The erosion of an image $f(x, y)$ by a flat SE $b(x, y)$ is defined as

$$[A \ominus B](x, y) = min\{f(x + s, y + t) | (s, t) \in D_b\} \qquad (8.3)$$

where D_b is called the domain of b.

- For a nonflat SE, $b_N(x, y)$, equation (8.3) becomes

$$[A \ominus B](x, y) = min\{f(x + s, y + t) - b_N(s, t) | (s, t) \in D_b\} \qquad (8.4)$$

EXAMPLE 8.8

- Figure 8.16 shows examples of grayscale erosion and dilation with a nonflat ball-shaped structuring element with radius 5 (*se = strel('ball',5,5);*).

(a) (b) (c)

FIGURE 13.16 Grayscale erosion and dilation with a nonflat ball-shaped structuring element with radius 5: (a) input image; (b) result of dilation; (c) result of erosion.

8.7.2 OPENING AND CLOSING

- Grayscale opening and closing are as defined exactly as they were for binary morphology.
- The opening of grayscale image $f(x, y)$ by SE $b(x, y)$ is given by

$$f \circ b = (f \ominus b) \oplus b$$

- The closing of grayscale image $f(x, y)$ by SE $b(x, y)$ is given by

$$f \bullet b = (f \oplus b) \ominus b$$

- Grayscale opening and closing are used in combination for the purpose of smoothing and noise reduction, similar to what we saw in Section 8.5 for binary images.

EXAMPLE 8.9

- Figure 8.17 shows examples of applying grayscale opening and closing with a flat disk-shaped structuring element with radius 3 (*se = strel('disk',3);*) for noise reduction.
- The upper part of the figure refers to an input image corrupted by Gaussian noise of mean 0 and variance 0.01, whereas the bottom part refers to an input image corrupted by salt and pepper noise.

EXAMPLE 8.9

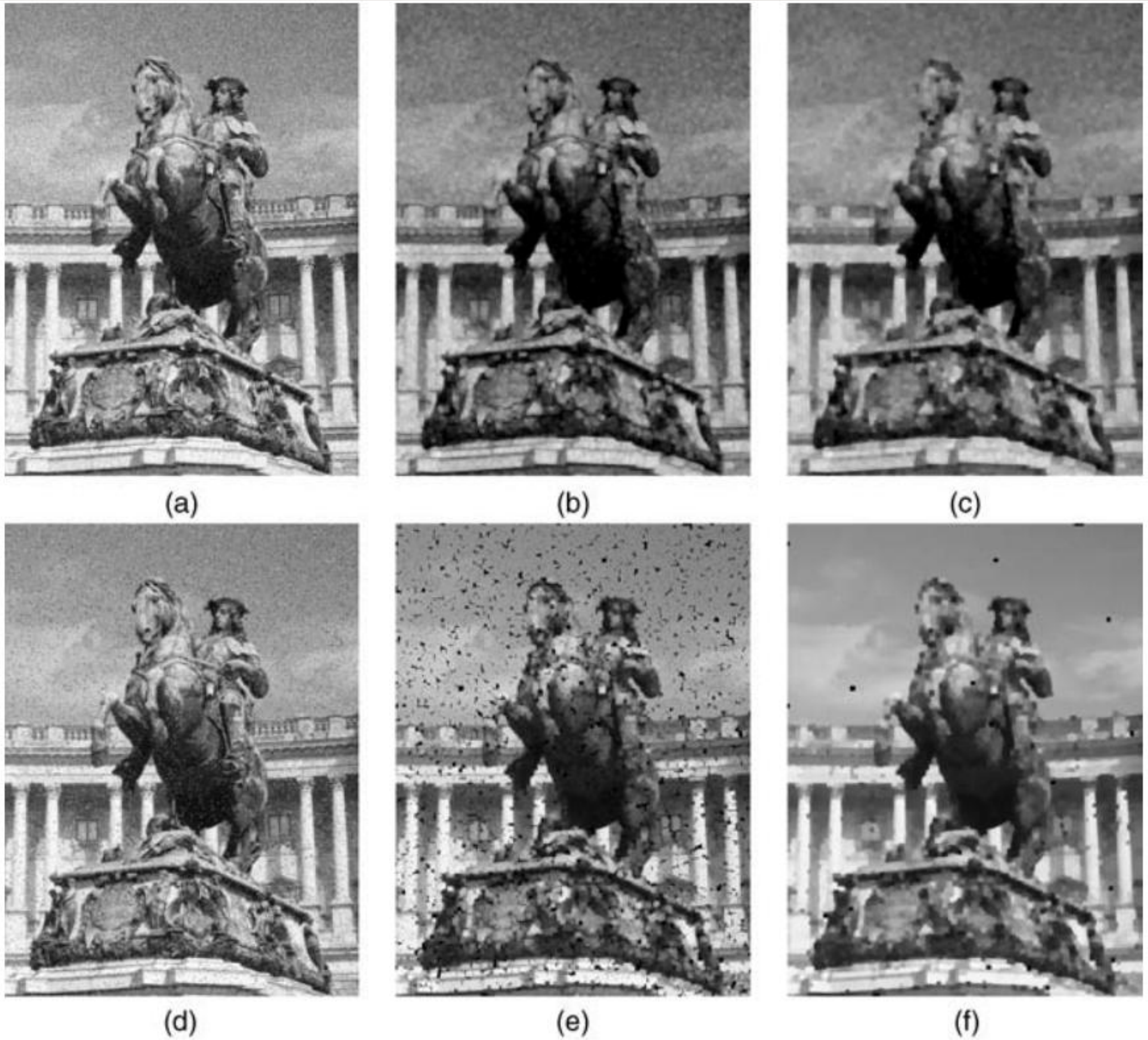


FIGURE 13.17 Grayscale opening and closing with a flat disk-shaped structuring element with radius 3: (a) input image (Gaussian noise); (b) result of opening image (a); (c) result of closing image (b); (d) input image (salt and pepper noise); (e) result of opening image (d); (f) result of closing image (e).

8.7.3 TOP-HAT AND BOTTOM-HAT TRANSFORMATIONS

- The *top-hat* and *bottom-hat* transformations are operations that combine morphological openings and closings with image subtraction. The top-hat transformation is often used in the process of *shading correction*, which consists in compensating for nonuniform illumination of the scene. Top-hat and bottom-hat operations can be combined for contrast enhancement purposes.
- Mathematically, the top-hat transformation of a grayscale image f is defined as the subtraction of f from its opening with SE b:

$$Top - hat\ (f) = f - (f \circ b)$$

and the bottom-hat transformation of f is given by the subtraction of the closing of f with SE b from f:

$$Bottom - hat\ (f) = (f \bullet b) - f$$

8.7.3 TOP-HAT AND BOTTOM-HAT TRANSFORMATIONS

In MATLAB

- IPT functions *imtophat* and *imbothat* implement top-hat and bottom-hat filtering, respectively.

EXAMPLE 8.10

- Figure 8.18 shows an example of applying *imtophat* and *imbothat* to improve the contrast of an input image I using a flat disk-shaped SE of radius 3. The output image was obtained as follows:

J = imsubtract(imadd(I,imtophat(I,se)), imbothat(I,se));

EXAMPLE 8.10

FIGURE 13.18 Example of using top-hat and bottom-hat filtering for contrast improvement: (a) input image; (b) output image.

8.8 TUTORIAL: BINARY MORPHOLOGICAL IMAGE PROCESSING

Goal

- The goal of this tutorial is to learn how to implement the basic binary morphological operations in MATLAB.

Objectives

- Learn how to dilate an image using the *imdilate* function.
- Learn how to erode an image using the *imerode* function.
- Learn how to open an image using the *imopen* function.
- Learn how to close an image using the *imclose* function.
- Explore the hit-or-miss transformation using the *bwhitmiss* function.

8.8 TUTORIAL: BINARY MORPHOLOGICAL IMAGE PROCESSING

1. Load and display the ‘blobs’ test image.

```
I = imread('blobs.png');
figure, imshow(I), title('Original Image');
```

Dilation

2. Create a 3x3 structuring element with all coefficients equal to 1.

```
SE_1 = strel('square',3);
```

3. Perform dilation using the generated SE and display the results.

```
I_dil_1 = imdilate(I,SE_1);
figure, imshow(I_dil_1), title('Dilated with 3x3');
```

Question 1. What happened when we dilated the image with a square 3× 3 SE?

8.8 TUTORIAL: BINARY MORPHOLOGICAL IMAGE PROCESSING

Let us now see what happens when using a SE shape other than a square.

4. Create a 1 × 7 SE with all elements equal to 1 and dilate the image.

```
SE_2 = strel('rectangle', [1 7]);
I_dil_2 = imdilate(I, SE_2);
figure, imshow(I_dil_2), title('Dilated with 1x7');
```

Question 2. What is the difference in results between the dilation using the 3 × 3 SE and the 1 × 7 SE?

Question 3. How would the results change if we use a 7 × 1 SE? Verify your prediction.

Question 4. What other SE shapes does the strel function support?

8.8 TUTORIAL: BINARY MORPHOLOGICAL IMAGE PROCESSING

Erosion

The procedure for erosion is similar to that for dilation. First, we create a structuring element with the *strel* function, followed by eroding the image using the *imerode* function. We will use the same two SEs already created in the previous steps.

5. Erode the original image with a 3 × 3 structuring element and display the results.

```
I_ero_1 = imerode(I, SE_1);
figure, imshow(I), title('Original Image');
figure, imshow(I_ero_1), title('Eroded with 3x3');
```

8.8 TUTORIAL: BINARY MORPHOLOGICAL IMAGE PROCESSING

6. Erode the original image with a 1x7 structuring element.

```
I_ero_2 = imerode(I, SE_2);

figure, imshow(I_ero_2), title('Eroded with 1x7');
```

Question 5. What is the effect of eroding the image?

Question 6. How does the size and shape of the SE affect the results?

8.8 TUTORIAL: BINARY MORPHOLOGICAL IMAGE PROCESSING

Opening

7. Perform morphological opening on the original image using the square 3 × 3 SE created previously.

```
I_open_1 = imopen(I, SE_1);
figure, imshow(I), title('Original Image');
figure, imshow(I_open_1), title('Opening the image');
```

Question 7. What is the overall effect of opening a binary image?

8. Compare opening with eroding.

```
figure, subplot(2,2,1), imshow(I), title('Original Image');
subplot(2,2,2), imshow(I_ero_1), title('Result of Erosion');
subplot(2,2,3), imshow(I_open_1), title('Result of Opening (3x3)');
```

8.8 TUTORIAL: BINARY MORPHOLOGICAL IMAGE PROCESSING

Question 8. How do the results of opening and erosion compare?

9. Open the image with a 1 × 7 SE.

```
I_open_2 = imopen(I, SE_2);

subplot(2,2,4), imshow(I_open_2), title('Result of Opening (1x7)');
```

Question 9. How does the shape of the SE affect the result of opening?

8.8 TUTORIAL: BINARY MORPHOLOGICAL IMAGE PROCESSING

Closing

10. Create a square 5 × 5 SE and perform morphological closing on the image.

```
SE_3 = strel('square',5);
I_clo_1 = imclose(I, SE_3);
figure, imshow(I), title('Original Image');
figure, imshow(I_clo_1), title('Closing the image');
```

Question 10. What was the overall effect of closing this image?

8.8 TUTORIAL: BINARY MORPHOLOGICAL IMAGE PROCESSING

11. Compare closing with dilation.

figure, imshow(I), title('Original Image');

figure, imshow(I_dil_1), title('Dilating the image');

figure, imshow(I_clo_1), title('Closing the image');

Question 11. How does closing differ from dilation?

8.8 TUTORIAL: BINARY MORPHOLOGICAL IMAGE PROCESSING

Hit-or-Miss Transformation

The HoM transformation is implemented through the *bwhitmiss* function, which takes three variables: an image and two structuring elements. Technically, the operation will keep pixels whose neighbors match the first structuring element, and will also keep any pixels whose neighbors do not match the second structuring element. This justifies the name of the operations: a hit for the first structuring element or a miss for the second structuring element.

Normally we would use *strel* to generate the structuring element; but in this case, we are not simply generating a matrix of 1s. Therefore, we will define the structuring elements manually.

12. Close any open windows.

8.8 TUTORIAL: BINARY MORPHOLOGICAL IMAGE PROCESSING

13. Define the two structuring elements.

SE1 = [0 0 0 0 0
0 0 0 0 0
0 1 1 0 0
0 0 1 0 0
0 0 0 0 0]

SE2 = [0 0 0 0 0
1 1 1 1 0
0 0 0 1 0
0 0 0 1 0
0 0 0 1 0]

8.8 TUTORIAL: BINARY MORPHOLOGICAL IMAGE PROCESSING

14. Apply the HoM operation on the original image.

```
I_hm = bwhitmiss(I,SE1,SE2);
figure, imshow(I), title('Original Image');
figure, imshow(I_hm), title('Hit-or-miss operation');
```

Question 12. What was the result of applying the HoM operation to the image with the given structuring elements?

8.8 TUTORIAL: BINARY MORPHOLOGICAL IMAGE PROCESSING

Question 13. How could we define the structuring elements so that the bottom left corner pixels of objects are shown in the result?

The *bwhitmiss* function offers a shortcut for creating structuring elements like the one above. Because the two structuring elements above do not have any values in common, we can actually use one array to represent both structuring elements.

8.8 TUTORIAL: BINARY MORPHOLOGICAL IMAGE PROCESSING

In the array in Figure 8.19, 1 refers to ones in the first structuring element, −1 refers to ones in the second structuring element, and all 0s are ignored. This array is called an interval.

15. Define an equivalent interval to that of the two structuring elements previously defined.

$$\text{Interval} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ -1 & -1 & -1 & -1 & 0 \\ 0 & 1 & 1 & -1 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & -1 & 0 \end{bmatrix}$$

8.8 TUTORIAL: BINARY MORPHOLOGICAL IMAGE PROCESSING

$$Interval = \begin{array}{|c|c|c|c|c|} \hline 0 & 0 & 0 & 0 & 0 \\ \hline -1 & -1 & -1 & -1 & 0 \\ \hline 0 & 1 & 1 & -1 & 0 \\ \hline 0 & 0 & 1 & -1 & 0 \\ \hline 0 & 0 & 0 & -1 & 0 \\ \hline \end{array}$$

FIGURE 13.19 Combining two structuring elements into one for the HoM transformation.

```
I_hm2 = bwhitmiss(I,interval);
figure, imshow(I_hm), title('Using two SEs');
figure, imshow(I_hm2), title('Using interval');
```

Question 14. Will this technique work if the two structuring elements have elements in common? Explain.

8.9 TUTORIAL: BASIC MORPHOLOGICAL ALGORITHMS

Goals

- The goal of this tutorial is to learn how to implement basic morphological algorithms in MATLAB.

Objectives

- Learn how to perform boundary extraction.
- Explore the *bwperim* function.
- Learn how to fill object holes using the *imfill* function.
- Explore object selection using the *bwselect* function.
- Learn how to label objects in a binary image using the *bwlabel* function.
- Explore the *bwmorph* function to perform thinning, thickening, and skeletonization.

8.9 TUTORIAL: BASIC MORPHOLOGICAL ALGORITHMS

Boundary Extraction

1. Load and display the test image.

```
I = imread('morph.bmp');
figure, imshow(I), title('Original image');
```

2. Subtract the original image from its eroded version to get the boundary image.

```
se = strel('square',3);
I_ero = imerode(I,se);
I_bou = imsubtract(I,I_ero);
figure, imshow(I_bou), title('Boundary Extraction');
```

8.9 TUTORIAL: BASIC MORPHOLOGICAL ALGORITHMS

3. Perform boundary extraction using the bwperim function.

```
I_perim = bwperim(I,8);
figure, imshow(I_perim), title('Boundary using bwperim');
```

Question 1. Show that the I_perim image is exactly the same as I_bou.

Question 2. If we specify 4-connectivity in the bwperim function call, how will this affect the output image?

8.9 TUTORIAL: BASIC MORPHOLOGICAL ALGORITHMS

Region Filling

We can use the imfill function to fill holes within objects (among other operations).

4. Close any open figures.

5. Fill holes in the image using the imfill function.

```
I_fill1 = imfill(I,'holes');
figure, imshow(I_fill1), title('Holes filled');
```

Function imfill can also be used in an interactive mode.

8.9 TUTORIAL: BASIC MORPHOLOGICAL ALGORITHMS

6. Pick two of the three holes interactively by executing this statement. After selecting the points, press Enter.

```
I_fill2 = imfill(I);
imshow(I_fill2), title('Interactive fill');
```

Question 3. What other output parameters can be specified when using imfill?

8.9 TUTORIAL: BASIC MORPHOLOGICAL ALGORITHMS

Selecting and Labeling Objects

The *bwselect* function allows the user to interactively select connected components—which often correspond to objects of interest—in a binary image.

7. Close any open figures.

8. Select any of the white objects and press Enter.

```
bwselect(I);
```

Question 4. In the last step, we did not save the output image into a workspace variable. Does this function allow us to do this? If so, what is the syntax? In many cases, we need to label the connected components so that we can reference them individually. We can use the function bwlabel for this purpose.

8.9 TUTORIAL: BASIC MORPHOLOGICAL ALGORITHMS

9. Label the objects in an image using bwlabel.

```
I_label = bwlabel(I);
figure, imshow(I_label,[]), title('Labeled image');
```

Question 5. What do the different shades of gray represent when the image is displayed?

Question 6. What other display options can you choose to make it easier to tell the different labeled regions apart?

8.9 TUTORIAL: BASIC MORPHOLOGICAL ALGORITHMS

Thinning

Thinning is one of the many operations that can be achieved through the use of the *bwmorph* function (see Table 8.1 for a complete list).

10. Use bwmorph to thin the original image with five iterations.

```
I_thin = bwmorph(I,'thin',5);
figure, imshow(I_thin), title('Thinning, 5 iterations');
```

Question 7. What happens if we specify 10, 15, or Inf (infinitely many) iterations instead?

8.9 TUTORIAL: BASIC MORPHOLOGICAL ALGORITHMS

Thickening

11. Thicken the original image with five iterations.

```
I_thick = bwmorph(I,'thicken',5);
figure, imshow(I_thick), title('Thicken, 5 iterations');
```

Question 8. What happens when we specify a higher number of iterations?

Question 9. How does MATLAB know when to stop thickening an object?

(Hint: Check to see what happens if we use Inf (infinitely many) iterations?

8.9 TUTORIAL: BASIC MORPHOLOGICAL ALGORITHMS

Skeletonization

12. Close any open figures.

13. Generate the skeleton using the bwmorph function.

```
I_skel = bwmorph(I,'skel',Inf);
figure, imshow(I_skel), title('Skeleton of image');
```

Question 10. How does skeletonization compare with thinning? Explain.

WHAT HAVE WE LEARNED?

- Mathematical morphology is a branch of image processing that has been successfully used to represent, describe, and analyze shapes in images. It offers useful tools for extracting image components, such as boundaries and skeletons, as well as pre- or postprocessing images containing shapes of interest.
- The main morphological operations are erosion (implemented by the *imerode* function in MATLAB), dilation (*imdilate*), opening (*imopen*), closing (*imclose*), and the hit-or-miss transform (*bwhitmiss*).

WHAT HAVE WE LEARNED?

- The structuring element is the basic neighborhood structure associated with morphological image operations. It is usually represented as a small matrix, whose shape and size impact the results of applying a certain morphological operator to an image.
- Morphological operators can be combined into useful algorithms for commonly used image processing operations, for example, boundary extraction, region filling, and extraction (and labeling) of connected components within an image.

PROBLEMS

Problem 1. Use *strel* and *getsequence* to create SEs of different shape and size and inspect their decomposed structuring elements.

Problem 2. Write a MATLAB script (or function) that extracts the connected components of a binary image and displays the results using different colors for each component and overlays a cross-shaped symbol on top of each component's center of gravity.

Problem 3. How many connected components would the *bwlabel* function find, if presented with the negative (i.e., the image obtained by swapping white and black pixels) of the image in Figure 2.8a as an input image? Explain.

END OF LECTURE 8

LECTURE 9 – EDGE DETECTION